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Towards Inclusion of the Personality Framework: A Systematic Review of AI-driven Mindful Eating Interventions for Adaptive and Personalized Eating Behaviors

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The Editor-in-Chief,
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(TiiS).

Sir,

Manuscript Submission

With respect to the call for submissions to the special issue "*Systematizing Knowledge in Interactive Intelligent Technologies*" in your journal, ACM Transactions on Interactive Intelligent Systems, we are delighted to submit our manuscript titled "*Towards Inclusion of the Personality Framework: A Systematic Review of AI-driven Mindful Eating Interventions for Adaptive and Personalized Eating Behaviors*" to this special issue and journal of interest.

This manuscript covers review findings collected from the synthesis of 46 papers that we synthesized from databases such as IEEE Xplore, ACM Digital Library, PubMed, Scopus, ScienceDirect, and CINAHL. Our findings show that 18 out of 20 interventions providing mindfulness components were considered effective, leveraging diverse user interface strategies, such as personalized feedback dashboards, meal journaling screens, and social or gamified features to enhance engagement and self-reflection. Furthermore, results show an overall small to moderate effect size of these interventions for managing. Overall, we proposed three design recommendations (D1-D3) to guide future interventions. These recommendations include developing ethically aware (D1), adaptive systems; incorporating Big Five personality dimensions to enhance personalization (D2); and advancing multisensory and immersive technologies to enhance user engagement (D3).

We declare that this research is original and has not been submitted to other venues prior to this submission. In addition, we declare that no financial support was obtained for this study.

Thank you for the privilege of submitting this manuscript. We look forward to hearing from you soon.

Best Regards,
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Towards Inclusion of the Personality Framework: A Systematic Review of AI-driven Mindful Eating Interventions for Adaptive and Personalized Eating Behaviors

Anonymized Authors*

Mindful eating interventions have gained increasing attention as effective strategies for promoting healthy eating behaviors and improving overall well-being. However, the integration of adaptive and personalized approaches, particularly those informed by personality traits such as the Big Five, remains underexplored. This systematic review synthesizes findings from 48 studies, examining technological interventions ranging from mobile applications to wearable and multisensory systems. We followed PRISMA guidelines and employed the SALSA framework for structured search, appraisal, synthesis, and analysis across six interdisciplinary databases. Our findings indicate that 18 out of 20 interventions providing mindfulness components were considered effective, leveraging diverse user interface strategies, such as personalized feedback dashboards, meal journaling screens, and social or gamified features to enhance engagement and self-reflection. Furthermore, results show an overall small to moderate effect size of these interventions for managing different eating behaviors. We found significant limitations, including adult-centric focus, exclusion of individuals with mental health or dietary conditions, and minimal integration of personality-driven tailoring, and we proposed three design recommendations (D1-D3) to guide future interventions. These recommendations include developing ethically aware (D1), adaptive systems; incorporating Big Five personality dimensions to enhance personalization (D2); and advancing multisensory and immersive technologies to enhance user engagement (D3). This review contributes to human-computer interaction (HCI) and artificial intelligence (AI) research by offering a framework for designing future adaptive, personality-informed, mindful eating technologies that support diverse user needs and promote sustained healthy eating habits.

CCS CONCEPTS • Human-centered computing → Human-computer interaction (HCI).

Additional Keywords and Phrases: Adaptive behavioral interventions, Artificial intelligence (AI), Mindful eating, Personalized nutrition, Personality

ACM Reference Format:

First Author's Name, Initials, and Last Name, Second Author's Name, Initials, and Last Name, and Third Author's Name, Initials, and Last Name. 2018. The Title of the Paper: ACM Conference Proceedings Manuscript Submission Template: This is the subtitle of the paper, this document both explains and embodies the submission format for authors using Word. In Woodstock '18: ACM Symposium on Neural Gaze Detection, June 03–05, 2018, Woodstock, NY. ACM, New York, NY, USA, 10 pages. NOTE: This block will be automatically generated when manuscripts are processed after acceptance.

1 INTRODUCTION

Over the past decade, artificial intelligence (AI)-driven mindful eating interventions have gained increasing attention as a means to promote healthier eating behaviors and improve overall well-being. These interventions emphasize awareness and presence during food consumption, leveraging AI to deliver personalized, adaptive dietary guidance, due

* Place the footnote text for the author (if applicable) here.

to the complexity of human eating behaviors. AI-powered applications now enable the analysis of eating habits, real-time tracking of food intake, and the provision of tailored meal planning, making mindful eating more accessible and effective for diverse populations. By harnessing sophisticated algorithms, these technologies assess individual nutritional needs, preferences, and behavioral patterns, offering real-time feedback and recommendations that support sustained healthy eating behaviors.

Although technological advancements have enhanced the development of mindful eating interventions, a critical dimension that remains underexplored is the role of individual personality traits in shaping eating behaviors and influencing responses to dietary interventions. Personality traits frameworks like the Big-Five Factor Model provide an organized and research-driven way of describing the vast variety of individual differences in behavior, thinking, and emotional functioning. Another set of personality frameworks, such as Eysenck's Three-Factor Model or the Myers-Briggs Type Indicator (MTBI), has attempted to describe human personality; however, they do not produce the same empirical support and practicality across cultures compared to the Big Five [3,33]. The Big Five's strength is related to the broad classifications it represents, its consistency in reproducing structural factors across a variety of populations, and predictive power with regard to factors related to human experience, such as job performance, relationship satisfaction, and mental health.

These frameworks have informed user behavior modeling in health-related interventions over the years [58,70,93], including healthy eating interventions [5]. Research has established an empirical relationship between eating behaviors and personality [38,44,55,91]. Additionally, studies have shown that personality characteristics—such as conscientiousness, neuroticism, openness to experience, and extraversion—are closely linked to dietary choices, self-regulation, and susceptibility to emotional eating. For instance, individuals who score high on conscientiousness tend to make healthier food choices (e.g., fruit consumption) and maintain better dietary discipline [65], while those who score high on neuroticism struggle with increased disordered eating and portion control [45]. These findings suggest that a one-size-fits-all approach to mindful eating interventions may overlook the nuanced ways in which personality influences eating patterns and intervention outcomes. Nevertheless, gaps remain in existing literature, particularly within the human-computer interaction (HCI) community, regarding how AI-driven nutrition systems are designed to interact with users in behavior-sensitive, emotionally aware, and adaptive ways. There is limited understanding of how eating behaviors and personality traits inform the design and development of adaptive, intelligent interfaces for mindful eating. Furthermore, the extent to which AI-powered nutrition technologies support interactive, user-centered experiences that promote mindful eating, through personalized interface design, user engagement, and behavioral feedback, remains underexplored.

Several reviews have explored AI-driven nutrition systems, but little or no reviews have comprehensively connected eating behavior patterns, personality traits, and adaptive feedback design in AI-driven nutrition technologies. Notable, yet partial reviews in the last five years lack coverage of granular eating behaviors and their integration into adaptive, interactive systems for personalized nutrition. For example, Lopez-Barreiro et al. [73] analyzed 34 studies published up to August 2024, focusing on AI and machine learning-based recommendation systems across diet, physical activity, and wellness domains. While it offered insights into algorithmic approaches and data types used, it revealed a notable gap in how eating behavior patterns, personality traits, and adaptive feedback design are modeled or utilized in AI-driven nutrition systems, particularly within the scope of AI-HCI integration for delivering personalized, interactive, and behavior-sensitive dietary support. In another related review, Casu et al. [23] analyzed 15 studies published up to April 2024, focusing on the use of AI chatbots across a range of mental health interventions. While the review identified promising outcomes in emotional well-being and behavior change, it revealed a significant gap in the modeling of eating behavior patterns, personality traits, and adaptive feedback design within AI-driven health systems, particularly in the

context of HCI, where personalization, engagement, and ethical human-AI interaction are crucial for delivering effective, behavior-sensitive nutritional guidance. Other related reviews with a critical gap in this domain we seek to explore include Zheng et al. [117], which synthesized AI applications for dietary assessment without addressing behavioral or personality adaptation, and Esposito et al. [38], which examined personality–diet associations without exploring their integration into AI-driven, user-centered intervention design.

With the rise of data-rich food tracking apps, LLM-driven food interfaces, and other emerging technologies such as wearable metabolic sensors and persuasive conversational agents [34,72,99,103], there exist increasing opportunities to transform the way people think about their diets, use and interact with nutritional information, as well as food decision-making more generally. Relatedly, it has also become critical to examine how behavior- and trait-aware AI systems are being designed and deployed. Motivated by these gaps, this systematic review seeks to synthesize current evidence on AI-driven mindful eating interventions, with a particular focus on the integration of personality frameworks for the promotion of adaptive and personalized eating behaviors. We will investigate how behavioral and psychological models are incorporated into the interface logic, feedback presentation, and interaction strategies of these systems. Specifically, we will examine how users interact with systems that adapt to fine-grained behaviors, such as water intake timing, eating position, utensil use, or post-meal activity, and how these features influence user experience and intervention outcomes over time. Through this comprehensive synthesis, we aim to identify pathways for the development of more effective, engaging, and sustainable digital health solutions that understand the complexity of human individuality. By bridging the gap between AI-driven personalization and psychological theory, this review will provide actionable insights for researchers, designers, and practitioners seeking to advance the field of mindful eating interventions and promote healthier, more adaptive eating behaviors for all technology users.

To guide this review, we proposed the following four research questions:

- RQ1: How are AI-driven mindful eating interventions tailored through user-interaction design and personality-based adaptations for diverse eating behaviors?
- RQ2: How do people engage with technologies providing mindful eating interventions for different eating behaviors?
- RQ3: How effective are mindful eating interventions provided for different eating behaviors?
- RQ4: What are the limitations and design opportunities towards making mindful eating interventions adaptive to the personality of users based on the Big-Five traits?

We seek to answer RQ1 by synthesizing data covering the different eating behaviors (e.g., fasting practices, timing between meals, cravings, eating with distractions), personality traits (e.g., agreeableness, openness to experience), types of technologies (e.g., wearable, mobile app), components of the user interface and their functionalities (e.g. login, activity screens), category of users (e.g., adults, children), age of the users (e.g., 18 – 30 years), features of AI in eating behavior technologies (e.g., ML algorithms, sensors), types of ML algorithms (e.g. CNN), types of sensors, and study locations (e.g. laboratory experiments and online study). To answer RQ2, we will analyze data such as specific name of the intervention, data collected using the intervention (e.g., based on user's characteristics), interaction modalities (sound, haptic, gesture, visual), interaction types, types of meal (e.g. breakfast), description of the type of mindful eating intervention provided, and class of intervention (e.g., adaptive/personalized intervention). In addition, we answer RQ3 by synthesizing data such as measures outcomes (e.g. eating speed), instruments used for data collection and evaluation, statistical tests/method of analysis (e.g., ANOVA), results of empirical evaluation (e.g., $p < 0.001$), effect size, accuracy of the ML algorithm (e.g., 90%), effectiveness (e.g., yes, no or maybe), and generalization of effectiveness based on exclusion criteria (e.g., what participants were excluded?). We answer RQ4 by synthesizing data covering the challenges and limitations of existing studies, prospective future work for developing mindful eating interventions for people with eating behaviors

identified and with the different personality traits, potential ethical issues, and implications of key findings as design opportunities for future work.

We conducted a literature search following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) procedure and guidelines [89]. Our search covered five databases (ACM Digital Library, IEEE Xplore, PubMed, Scopus, and Cinahl,), and we selected 46 papers that were valid for answering our research questions. This informed us of our data extraction and data analysis method using descriptive statistics, inferential statistics, and content analysis. Based on our data analysis, our findings show that most mindful eating interventions demonstrated positive behavioral impacts, with strong evidence supporting their effectiveness in reducing eating speed, enhancing food awareness, and promoting healthier dietary choices. Moreover, trends in user interface designs such as real-time feedback, activity screens, and social features played a significant role in improving user engagement and adherence. Overall, we propose three design recommendations: D1: integrate adaptive, personality-driven feedback systems to tailor interventions to individual traits and behaviors; D2: adopt ethics-by-design approaches to ensure user privacy, transparency, and inclusivity; and D3: incorporate multisensory and immersive interface strategies to deepen engagement and enhance the experiential quality of eating interventions. These recommendations highlight opportunities for more personalized, effective, and ethically responsible intervention designs.

Based on our findings, this review contributes to the growing body of knowledge in the HCI and AI research communities in three ways: (1) we uncover different trends that shape the development of mindful eating interventions and illustrate their technological and design evolution, (2) we provide design frameworks that will guide the creation of adaptive and personalized intervention systems grounded in personality science, and (3) we foster interdisciplinary integration between behavioral science, HCI, and AI-driven health technologies. Our systematic review will facilitate the modeling of personality traits frameworks with advanced sensing and AI-driven personalization strategies, thereby enhancing healthy eating behaviors and quality of life for diverse user populations. Additionally, these insights will inform the next generation of mindful, human-food interaction, autonomous, and adaptive eating technologies.

2 METHODOLOGY

We conducted a systematic literature review (SLR) following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [89], with search protocols tailored to nutrition, artificial intelligence (AI), behavioral science, and human-computer interaction (HCI) domains. This methodology was chosen because: (1) it supports a structured and transparent synthesis of interdisciplinary studies spanning technical system development, behavior change theories, and user-centered design practices; (2) it enables rigorous comparison of AI personalization strategies, while addressing interface-level interaction strategies, including user engagement patterns and usability considerations; and (3) it provides a strong foundation for proposing a conceptual framework for adaptive mindful eating systems that integrate both personality trait and behavioral dimensions. Besides, it has become one of the widely used approaches in most HCI systematic review studies (e.g., [6,24,95])

To guide our analysis and reporting, we employed Grant and Booth's framework: Search, Appraisal, Synthesis, and Analysis (SALSA) methodology [17]. This process provides a clear structure for organizing our findings: (1) Search helps us to report relevant sources we used to retrieve our articles; (2) Appraisal helps us to critically report the quality and relevance of the studies we included in our review; (3) Synthesis helps us to present our findings across studies to identify patterns and gaps; and (4) Analysis helps us to conclude to inform future research.

2.1 Search

To gather relevant articles for our systematic review, we employed comprehensive search keywords:

("*eating behavior*" OR "*dietary patterns*") AND ("*personality traits*")
AND ("*artificial intelligence*" OR "*AI*" OR "*app*")

Using these search keywords, we searched six major databases: ACM Digital Library, IEEE Xplore, ScienceDirect, PubMed, Scopus, and Cinahl. These databases were selected due to their interdisciplinary coverage of healthy eating, technology development, behavioral science, and artificial intelligence. Furthermore, they have been widely used in existing related reviews, ensuring comprehensive coverage of existing literature across related domains (e.g., [25,74]). In other words, this inclusion enhanced the synthesis of diverse perspectives and uncovered emerging patterns critical to understanding mindful eating interventions in technology-supported contexts. To achieve this, we conducted a comprehensive search targeting titles, abstracts, and full texts that contained our predefined keywords. This approach returned a total of 72,199 papers distributed as follows: ACM Digital Library (3,134), IEEE Xplore (217), PubMed (122), ScienceDirect (68,267), Scopus (451), and Cinahl (8). The search was performed on April 30, 2025.

2.2 Appraisal

We converted the search results from each of the databases into BibTeX format and then uploaded them to Rayyan (https://www.rayyan.ai/). Rayyan is a free web and mobile application that supports systematic reviews by enabling a semi-automated screening process and facilitating collaboration among reviewers [88]. This tool has been widely adopted in the HCI community for systematic reviews due to its efficiency in handling large datasets and its robust tagging and filtering features [1]. Thus, we used Rayyan to screen and remove duplicate papers, resulting in the exclusion of 11 duplicates. Next, we performed a title and abstract screening using predefined keywords, such as mindfulness, eating, behavior, intervention, and personalization, which led to the removal of 63,825 irrelevant papers. After this screening stage, 8,215 papers remained potentially relevant for full-text review. Our final inclusion of papers was guided by the following criteria (Table 1):

Table 1: Inclusion/exclusion criteria used for our article selection process.

S/N	Inclusion Criteria	Exclusion Criteria	Number of articles excluded (n)
1.	Empirical studies with human subjects.	Empirical studies without human subjects.	3
2.	Studies related to mindful eating.	Studies not related to mindful eating.	3651
3.	Original research.	Review papers, opinion pieces, and editorials without original empirical data.	4261
4.	Involved a kind of Technology/AI models	No Technology or AI models	254

Based on these exclusion criteria, we excluded a total of 8169, leaving 46 valid articles in preparation for our data analysis phase. To ensure the integrity of our systematic review, all six researchers took part in the screening and inclusion process for thorough selection and cross-validation of eligible articles. All six researchers rated papers uploaded to Rayyan and independently selected papers that were eligible based on our outlined exclusion criteria. After this first step, all six researchers converged to cross-validate the selected papers. We computed the interrater reliability ratings [30] to assess the reliability of our ratings. Our findings show strong agreement among all six researchers with a Fleiss' kappa of 0.78. According to Cole [31]Kappa values between 0.61 and 0.80 indicate substantial agreement. Our result demonstrates a high level of consistency among reviewers, reinforcing the rigor of our screening process [43]. Figure 1 presents our PRSIMA approach, showing the flow in our selection process

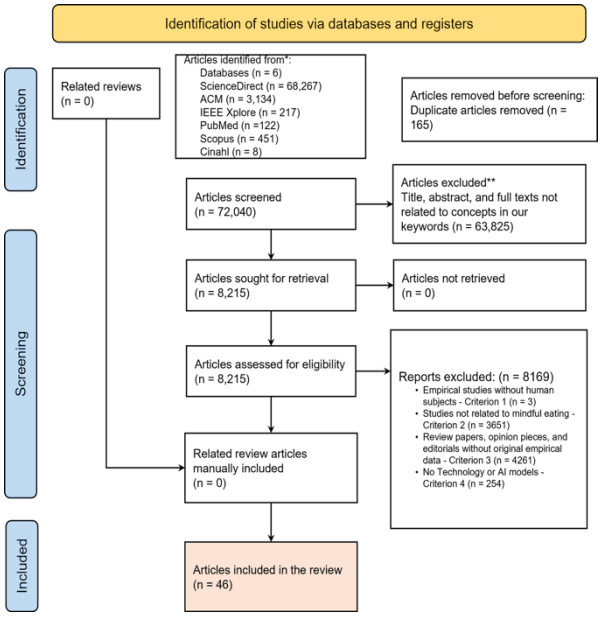

Figure 1: Adapted PRISMA diagram [89] showing our scoping review search and filtering processes. The asterisk symbols are typically used to depict the order of searches conducted, based on the number of search methods found [40].

2.3 Synthesis

After we completed the inclusion and exclusion procedure, we developed a comprehensive coding sheet to systematically extract relevant information from the selected papers (Appendix A.2). Specifically, we extracted data related to key codes that align with our research objectives and facilitate cross-study comparisons. These codes included: study title, type of technology used (e.g., mobile app, wearable, multisensory system), targeted eating behaviors, intervention components (e.g., mindful eating prompts, feedback mechanisms), user interface design features, measurement outcomes, instruments for data collection, exclusion criteria, and reported challenges and limitations. Additionally, we coded for aspects such as integration of personality-driven approaches, ethical considerations, and characteristics of the target user populations. Table 2 presents a cross-section of these codes and illustrates how they correspond to our predefined research questions (RQ1–RQ4). We provide a comprehensive table showing all our codes in Appendix A.1. Our data extraction using these codes ensured consistency and facilitated the synthesis of our findings.

Table 2: Cross-section of codes, descriptions, and research questions associated with them.

S/N	Code	Description	Research Questions
1.	Bibliometric data	This covered information about the article's publication year, type of publication, venue, DOI, and author keywords.	Bibliometric analysis
2.	Broad category of Eating Behavior (e.g. fasting practices, timing between meals, cravings, eating with distractions)	Main eating behaviors targeted, such as fasting practices, timing between meals, cravings, or eating with distractions.	RQ2, RQ3, and RQ4
3.	Personality Traits	Personality dimensions considered in the study (e.g., Big Five traits) and how they relate to eating behaviors.	RQ2 and RQ5

S/N	Code	Description	Research Questions
4.	Type of technology (wearable, mobile app)	The technological platform used includes mobile apps, wearables, VR, or multimodal systems.	RQ1
5.	Components of the user interface with an explanation of their functionalities (e.g., login, activity screens)	Description of UI elements and functionalities (e.g., login, activity screens, meal diaries, feedback dashboards).	RQ1
6.	Category of the users of technology (e.g., adults, children)	The target demographic of the intervention (e.g., adults, children, families).	RQ5
7.	Age of the users of technology (e.g., 20 or age bin if not provided as a single instance)	Reported age or age bins of study participants.	RQ5
8.	Features of AI in eating behavior technologies (e.g., ML algorithms, sensors, Explainability and Interpretability with SHAP)	AI capabilities applied, including machine learning algorithms, sensor fusion, and explainability features (e.g., SHAP).	RQ3
9.	If ML algorithm state, the type (e.g., CNN)	Specific types of machine learning algorithms used (e.g., CNN, LSTM, decision tree).	RQ3, RQ4
10.	If sensors, please state the types of sensors	Sensors used in technology such as IMUs, cameras, audio, piezoelectric, or biosensors.	RQ3

2.4 Analysis

To analyze the data, we extracted it into a coding sheet based on Table 1, and we cleaned the sheet by removing irrelevant entries and standardizing the format. First, we carefully removed data that did not align with our research questions and summarized the entries where additional text added ambiguity. Additionally, we standardized our coding sheet to handle missing data points by reconfirming entries from the specific papers involved. These rigorous cleaning procedures provided a quality dataset for our analysis. Next, we performed descriptive analyses and content analyses to evaluate patterns, frequency counts, and trends across focal themes, such as types of interventions, technological approaches, target behaviours, and personalization methods. For the descriptive analysis, we used Microsoft Excel to process and visualize data related to the bibliometric analysis, RQ1, RQ2, RQ3, RQ4, and RQ5. This software allows us to calculate basic statistics and create relevant charts and graphs. For the content analysis, we employed NVivo 14 [19], a qualitative data analysis software. NVivo was used to code and analyze textual data for answering research questions RQ5, RQ6, and parts of the data from RQ1. This software facilitated the identification of themes, patterns, and relationships within the qualitative data.

3 RESULTS

In this section, we present our results under two main categories: (1) bibliometric analysis, and (2) research questions. The bibliometric analysis summarizes publication trends and general characteristics of the included studies, detailed in Section 3.1. The subsequent sections address our research questions: Section 3.2 presents results for RQ1, exploring trends in tailoring mindful eating interventions for different eating behaviors; Section 3.3 addresses RQ2, examining how individuals engage with technologies providing these interventions; Section 3.4 reports on RQ3, investigating the effectiveness of the interventions; and Section 3.5 covers RQ4, discussing limitations and design opportunities for making mindful eating interventions adaptive to users' personalities based on the Big-Five traits.

3.1 Bibliometric Analysis

We analyzed various bibliometric data extracted from our coding sheet, including publication year, publication type (journal or conference), publication database (ACM or IEEE), author keywords, and geographical location of the studies.

This analysis aimed to highlight emerging patterns and the growing interest of the research community in mindful eating interventions across different regions. Understanding these patterns provides a foundation for interpreting our subsequent results and situating them within the broader research landscape.

Based on our analysis of the distribution of articles published within the past decade (2015–2025, Figure 2), we found that more articles emerged in recent years, particularly in 2022 (9 articles), compared to earlier years such as 2015 and 2016 (3 and 1 articles, respectively). This trend highlights the growing research interest in exploring mindful eating behavior.

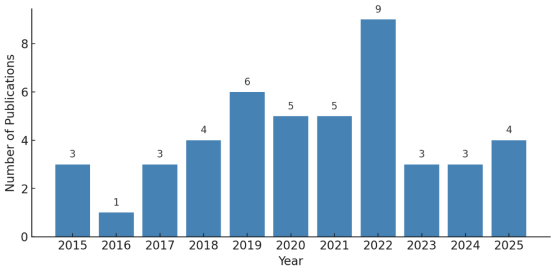

Figure 2: Distribution of a number of articles published from 2015 – 2025. This shows that more articles were published in the year 2022. Also, the year 2025 is showing a rising potential with 4 articles, given that we collected data early 2025 (April - May).

Further analysis on the design trends within the past decade (2015–2025) shows that publications on mindful eating technologies have steadily evolved from early wearable and mobile monitoring solutions toward more personalized, socially embedded, and ecologically conscious systems (Figure 3). Initial studies focused on basic eating detection and self-awareness (2015 – 2022), while recent years (2023 – 2025) have emphasized multimodal sensing, interactive feedback, and the integration of cultural and emotional dimensions. Notably, there has been an increase in gamification, hybrid sensing approaches, and interventions that consider both individual and environmental health impacts, highlighting a shift toward more holistic and user-centered designs.

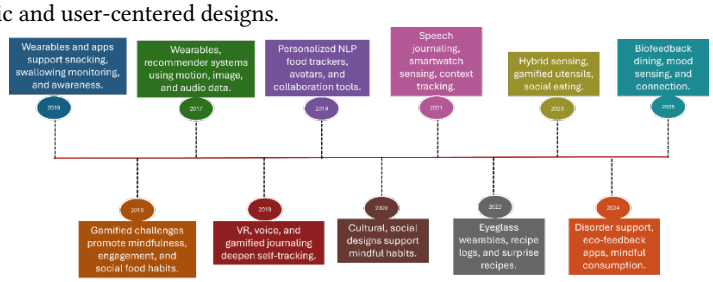

Figure 3: Timeline chart showing the advancement of research in this domain. This shows a recent shift to eating disorders, hybrid sensing, and gamified interventions

Additionally, we analyzed the distribution of articles published within the past decade (2015–2025) and indexed in different databases to illustrate the activity of various research communities (Figure 4). Findings show that more publications were indexed in the ACM database (38 articles, 79.2%), highlighting its strong focus on human-computer interaction (HCI) and psychological dimensions, including personality traits. In contrast, health-focused research is more prominent in PubMed and ScienceDirect (4 articles each, 8.3%), reflecting their emphasis on medical and clinical perspectives. Although IEEE had fewer publications (2 articles, 4.2%), its contributions typically emphasize adaptive systems and AI-driven approaches, demonstrating their impact from the technical and engineering perspectives. This underscores the need to balance the technological and psychological advancements emphasized in ACM with the health-

oriented evidence base in PubMed and ScienceDirect, alongside the AI-driven and adaptive approaches from IEEE, fostering more integrated and interdisciplinary collaborations. Overall, more articles were published in conference proceedings (27 articles, 56.2%) compared to journal articles (21 articles, 43.8%). This suggests that research in this area is often disseminated through conferences, which typically emphasize timely sharing of emerging findings within the research community. In contrast, journal publications, while fewer, also show a promising trend, with the number of comprehensive studies published in the last decade.

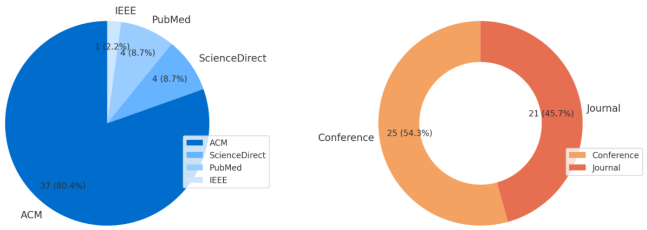

Figure 4: Distribution of articles by databases, conferences, and journals

Furthermore, our analysis of research coverage and interest across different geographical regions shows more publications in the USA (n = 17). Other countries with notable research coverage include China (n = 7), Japan (n = 7), Canada (n = 5), Australia (n = 4), and Mexico (n = 5). Moderate contributions were also observed from Germany (n = 4), South Korea (n = 3), Switzerland (n = 3), the United Kingdom (n = 3), and India (n = 3). Additional publications came from countries such as Singapore (n = 2), Colombia (n = 2), the Philippines (n = 2), the Netherlands (n = 2), Mongolia (n = 2), Paraguay (n = 2), and Italy (n = 2). Finally, smaller contributions were noted from New Zealand (n = 1), Denmark (n = 3), Spain (n = 1), France (n = 1), Israel (n = 1), Turkey (n = 1), Portugal (n = 1), and Belgium (n = 1). Our analysis was based on the number of occurrences with some publications having at least two countries. Figure 5 visualizes a cross-section of countries with a research presence in this domain, highlighting the geographical distribution and relative contribution of each country.

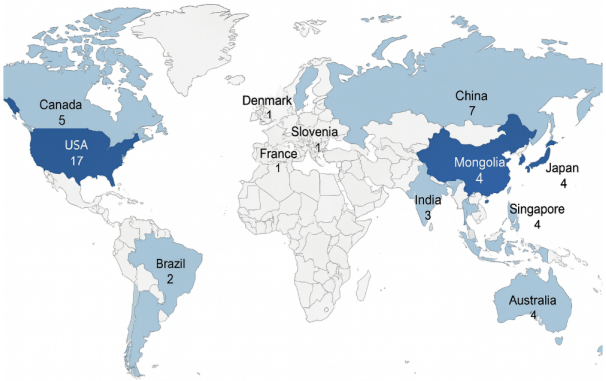

Figure 5: Geographical distribution showing the extent to which research has been covered in this domain

The analysis of keywords (Figure 6) from the provided dataset shows "eating" as the most frequently occurring concept, with 34 article references. This emphasizes the central focus on eating behaviors, practices, and interventions across various studies. Closely following is "food", appearing in 29 articles, reflecting the strong emphasis on dietary monitoring, nutrition, and food-related design interventions. "Behavior" (15 articles) and "health" (14 articles) further underscore the behavioral and health-oriented dimensions of this research domain, pointing to the critical intersection

of eating practices and overall well-being. "Mobile" (13 articles) and "sensing" (12 articles) illustrate the significant role of mobile and sensing technologies in capturing and influencing eating habits, emphasizing the importance of real-time, context-aware data collection.

Keywords such as "tracking" (11) and "monitoring" (8) reinforce this technological emphasis, highlighting how continuous and personalized feedback systems are core to these interventions. Lastly, "diet" and "interaction", each appearing in 9 articles, indicate a strong focus on dietary assessment and the interactive nature of human-food experiences, often facilitated through persuasive and participatory design approaches. Overall, these findings reveal a multidisciplinary focus, combining behavior change, mobile health (mHealth), sensing technologies, and interactive design to address challenges in eating habits and nutrition management. This suggests opportunities for future research to further integrate these domains, creating more comprehensive and adaptive interventions that support healthy eating and promote sustained behavior change.

Figure 6: Common keywords used by researchers in this domain, showing frequent use of terms such as eating food, mobile, behavior, health, and interaction

3.2 RQ1: What are the user-centered components for tailoring the user experience in mindful eating interventions, specifically in terms of technologies, user interaction, engagement, interface design, and feedback mechanisms?

To answer RQ1, we analyzed a combination of codes in our coding sheet. These include types of technologies (e.g., wearables, mobile apps), the name of the mindfulness intervention, user interface components, user demographics, Components of the user interface with explanation of their functionalities (e.g. login, activity screens), Interaction modalities (sound, haptic, gesture, visual), type of meal targeted (e.g., breakfast), and the broad category of eating behavior addressed (e.g., timing between meals, cravings, eating with distractions).

i. Technologies employed to deliver mindful eating interventions

Overall, we found that mobile apps were the most common technology type, used in 26 interventions, followed by wearables (11 interventions). Other combinations, such as multiplatform solutions, mobile app and wearable integrations, and others (not specified/other category covering tangible or those that were not explicitly mentioned) were less frequently reported (Figure 7).

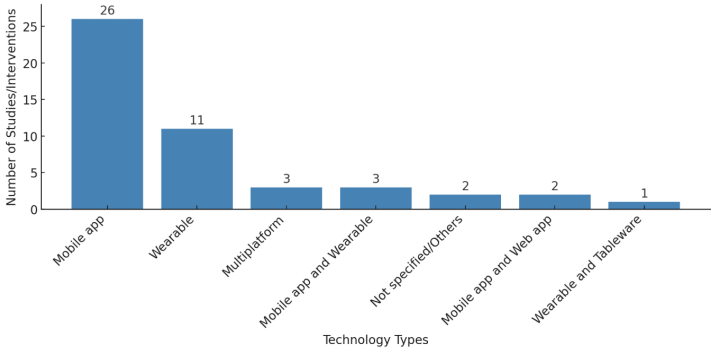

Figure 7: Distribution of different technological implementations of mindful eating interventions

Most studies leverage mobile apps and wearable technologies to provide real-time monitoring and adaptive feedback. For example, ViFeed strategically manipulates video playback to promote slow eating during screen-based dining [27], while FitNibble, an eyeglasses-based wearable, supports automatic diet monitoring and improves food journaling compliance through just-in-time notifications [12]. Other mobile applications, such as Q-Chef, use surprise-eliciting recipe recommendations to influence food choices and promote dietary diversity [48]. FoodCensor helps individuals with binge eating disorders by censoring digital food content and prompting reflection [28], whereas EcoMeal uses gamified eco-feedback via a virtual garden to raise awareness of environmental impacts [35]. Several studies incorporate multimodal sensing and passive data collection to adapt feedback based on context. For instance, Bites'n'Bits combines mobile and wearable data to infer meal types [14], while One More Bite? [81] and Sensing Eating Events in Context [11] leverage smartphone sensing and self-reports to understand consumption levels and social contexts. Similarly, DrinkWatch [39] and EatingTrak [115] utilize wearable sensors to detect fine-grained eating and drinking moments in everyday settings, enabling personalized nudges and self-monitoring. Other innovative systems include eat2pic [84], which transforms eating into a painting activity using sensor-equipped chopsticks to encourage slow and mindful eating, and Living Bento [26], which uses biofeedback to create an immersive and social dining experience through heartbeat-synchronized glowing noodles. Moreover, frameworks like "Design to Eat Smart!" [116] and studies such as "Co-Designing Food Trackers with Dietitians" [76] emphasize the importance of personalization and user-centered design in developing interventions that account for individual goals, cultural preferences, and environmental factors.

ii. User Interface and User Engagement

Among the emerging trends in tailoring mindful eating interventions identified in the studies we reviewed, we found seven key user interface (UI) design features that informed user engagement in adapting interventions across different eating behaviors: (1) login and profiles personalization, (2) activity screens and meal journaling interfaces, (3) feedback and visualization features, (4) reminder and notification systems, (5) social and gamified components, (6) integration of sensor-driven automation, and (7) emerging multisensory and immersive interfaces. These UI features enhance user engagement, facilitate self-reflection, and enable real-time feedback, collectively contributing to more personalized and effective behavior change support.

1. Login and profiles personalization

Most mobile-based interventions integrated login systems and profile creation features to enable user authentication, personalization, and secure data management. Examples include Snack Buddy [69], NutritionAvatar [30], Eat & Tell [2], EcoSant [51], Food4Thought [27], Nutrilize [70], Q-Chef [36], EcoMeal [25], RecipeLog [78], asken[40], Keenoa [14], and

calomeal [6]. This feature serves as a foundational design element not only for securing user access and ensuring data privacy but also for fostering personalized experience. This enables customized prompts, dietary recommendations, and progress tracking. Profiles can store user preferences, dietary restrictions, and even behavioral data, providing the groundwork for future adaptive or personality-driven intervention possibilities.

2. Activity screens and meal journaling interfaces

A central component across many mindful eating interventions is the activity or diary screen, which provides a dedicated space for users to log meals, snacks, mood states, portion sizes, and contextual factors such as social settings or emotional triggers [2,27,55–57]. These interfaces typically support a range of input modalities, including structured text fields, photo uploads (e.g., Keenoa [14]), barcode scanning, and voice recordings (e.g., FoodScrap's [55] reflective audio questions). This multimodal approach allows for richer, more personalized data capture, enabling users to articulate not just what they ate, but also why, when, and under what circumstances. In some cases, open-ended reflections or journaling prompts are used to deepen self-awareness and support emotional connection to eating habits, which are key principles in mindfulness-based interventions [69,70].

3. Feedback and visualization features

Feedback systems delivered via visual dashboards, nutrient breakdowns, eco-scores, gamified progress bars, or avatar-based coaching are widely utilized across mindful eating interventions to promote sustained engagement, reflective practice, and behavior change [10,25,30,69,70]. These systems convert abstract health-related or behavioral concepts into concrete, easy-to-understand visual formats. For instance, nutrient breakdown charts and portion visualizations help users comprehend macronutrient balance, calorie intake, and food diversity [6,14]. Similarly, eco-score dashboards (e.g., [25]) provide immediate feedback on the environmental impact of food choices, encouraging users to adopt more sustainable eating habits.

4. Reminder and notification systems

Reminders, both time-based and context-sensitive, are essential features seen in systems such as Crumbs [27], Eat & Tell [2], and ViFeed [20]. These notifications act as digital nudges, prompting users to slow down, check in with their hunger cues, or review daily logs. In several studies, adaptive reminders triggered by sensor data (e.g., detected bite speed or skipped meals) were explored, highlighting the move toward context-aware and just-in-time interventions [10,29].

5. Social and gamified components

To enhance motivation and foster a sense of community, many interventions incorporated social sharing options, family-based comparisons, or gamified challenge elements (e.g., [25,27,56,57,69]). These features introduce a competitive or collaborative dimension that engages users beyond individual self-monitoring. For example, family-based comparison tools offered in Snack Buddy [69] allow caregivers and adolescents to view and discuss each other's snack choices, fostering a supportive home environment and shared accountability. In EcoMeal [25], eco-score dashboards and gamified virtual gardens encourage users to compare environmental impacts and progress, adding an element of friendly competition that can drive behavior change. Similarly, Crumbs [27] uses daily food challenges and social group sharing (e.g., in private Facebook groups) to cultivate peer encouragement and reinforce commitment to mindful eating goals. These social features are particularly attractive to individuals with high levels of Extraversion, who are more motivated by external validation, group interactions, and social support [27,56,57,69], thereby limiting feelings of isolation while exhibiting different eating habit patterns.

6. Integration of sensor-driven automation

Wearable-integrated interventions, such as those using eyeglass-mounted sensors, wrist-mounted inertial measurement units (IMUs), or piezoelectric necklaces, aimed to reduce manual data entry and improve the accuracy of

real-time behavior tracking by passively detecting eating events and contextual signals [10,43,80]. These systems enable more seamless and unobtrusive monitoring, allowing users to engage in daily activities without constant app-based logging or manual input.

7. Emerging multisensory and immersive interfaces

A few innovative interventions moved beyond conventional screens, using ambient lighting, haptic feedback, and even taste and smell integration to shape and influence eating behaviors [19,31,47]. They stimulate multiple senses simultaneously, such as visual cues through glowing tableware, tactile signals via vibration or texture, or olfactory engagements, these interventions create more immersive, engaging experiences that enhanced users' awareness of their eating habits. This sensory-rich approach can help to create emotional connections with food, which may increase mindfulness during meals and enhance the perceived enjoyment of healthy eating practices.

iii. User Interaction based on Different Eating Behaviors and Feedback Mechanisms

User interaction is central to the effectiveness of mindful eating interventions. The integration of real-time monitoring, adaptive feedback, and multisensory interfaces reflects a trend toward context-aware, engaging, and user-centered mindful eating interventions. Our analysis revealed that different interventions were designed around specific eating behaviors, as summarized in Table 3. Eating with distractions was the most frequently targeted, with several systems aiming to reduce external stimuli and enhance attentional focus. Meal timing and rhythm were also commonly addressed, using time-based prompts and visual scheduling tools to support consistent eating habits. A number of studies focused on cravings, leveraging mood tracking, self-reflection prompts, or delay tactics to discourage impulsive eating. Eating speed was often regulated through bite or chew detection paired with real-time feedback, while snacking behaviors were addressed using journaling features and context-sensitive reminders. Additionally, some interventions emphasized nutrient intake visualization and personalized dietary tracking to enhance nutrition awareness and goal setting. These diverse behavioral targets informed the type and mode of user interaction across interventions.

Table 3: List of eating behaviors and references. This shows that more articles cover eating with distractions compared to other types of eating behaviors.

S/N	Eating Behaviors	Number of Studies	Selected References
1.	Eating with distractions	19	[9,11,12,14,18,26–28,37,42,51,57,64,66,75,76,81,82,104,116]
2.	Timing between meals / meal timing / meal rhythm	13	[9,11,12,37,51,56,59,75,78,81,86,98,115]
3.	Cravings	8	[9,11,12,28,48,51,81,96]
4.	Eating speed	5	[64,82,84,115,116]
5.	Snacking	4	[78,96,115,116]
6.	Dietary tracking / personalized nutrition / nutrient intake visualization	4	[8,18,76,97]

Most technologies leveraged visual modalities, often combined with other sensory feedback (haptic, auditory, gesture) to support real-time self-monitoring and promote awareness. Many interventions provided generic feedback aimed at fostering slower eating, reducing distractions, or increasing food-related self-reflection (Table 4). Meal type varies across interventions, ranging from general daily meals to specific contexts like snacking or social dinners. Physiological signal-driven interactions (e.g., chewing detection, hand-to-mouth motion, heartbeat feedback) facilitated

passive engagement and subtle behavioral cues without requiring active user input. Moreover, social and gamified elements, as employed in [9] and [13], promoted playful and community-oriented engagement. Overall, these findings highlight a growing trend in research towards the integration of personalized, adaptive, and multimodal interaction-driven designs in mindful eating interventions. Future interventions could benefit from incorporating more dynamic personalization strategies to enhance behavioral support and user engagement across diverse eating contexts.

Table 4: Different eating behaviors and interaction modalities showing how users interact with mindfulness interventions

S/N	Eating Behavior	Interaction Modes	Example Studies
1.	Eating with distractions	Visual prompts, sound alerts, minimal gesture input	[27,28,66]
2.	Slow eating and food awareness	Visual, gesture, camera feedback	[27,84]
3.	Emotional and mood-related eating	Visual interfaces, passive sensing	[10]
4.	Snacking, cravings, portion control	Visual, touch, haptic feedback	[96,116]
5.	Social or ritualized eating	Visual, social, embodied interaction	[9,13]
6.	Timing between meals	Visual, vibration prompts, gesture	[12,86]
7.	Chewing, swallowing monitoring	Audio-based detection, passive visual logs	[60,101]

Overall, we found 22 studies that offered mindful eating interventions in association with the different eating behaviors we identified in Section 3.2 (Figure 8). Interventions provided range from simple, generic prompts aimed at encouraging self-reflection (e.g., FoodCensor [28]) to highly adaptive and personalized systems that leverage real-time data and user characteristics (e.g., MAMAS [57], ViFeed [27], NutritionAvatar [41]). Many interventions emphasize raising awareness through visual feedback, guided questions, or nudges, supporting behaviors such as slowing eating pace, reducing distractions, or choosing healthier options [76,96,116]. While some systems, such as "Snack Buddy" [96] and "Design to Eat Smart!" [116], deliver tailored feedback based on detected eating patterns or contextual data, others rely on static or uniform guidance irrespective of individual differences (e.g., Crumbs [37], FoodScrap [75], Eat & Tell [2]). Despite the increasing focus on personalization, a considerable number of interventions still offer one-size-fits-all recommendations, indicating room for further advancement in user-centered adaptive strategies [2,28,37].

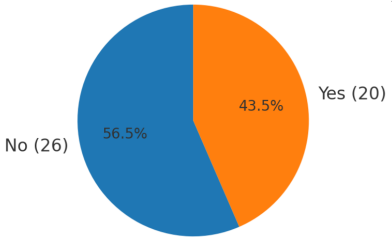

Figure 8: Distribution of the number of mindfulness eating interventions in our review

3.3 RQ2: How have *personality traits and eating behaviors* been employed to provide adaptive/personalized mindful eating interventions?

To answer RQ2, we analyzed three categories of code from our coding sheet. These include "Broad category of Eating Behavior (e.g., fasting practices, timing between meals, cravings, eating with distractions)", "Personality Traits, Description on the type of mindful eating intervention provided", and "Type of intervention: Adaptive/personalized intervention (yes/no)?" Findings show that interventions were averagely limited to using eating behaviors to inform adaptive or personalized recommendations informed by individual data (e.g., [10,113], Figure 9). Out of 36 eating behaviors we identified in Section 3.2, studies have provided adaptive/personalized interventions (25, 56.8%). For example, some studies tailored adaptive/personalized interventions to eating behaviors such as eating distractions (e.g., [11,18,26,57,82]), Snacking (e.g., [96,116]), Eating autonomy or picky eating [57], and Eating speed (e.g., [64,84,116]).

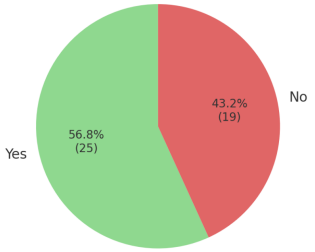

Figure 9: Studies that provided adaptive/personalized interventions

Table 5 shows those eating behaviors that have fully adaptive/personalized interventions compared to those with partial interventions.

Table 5: List of studies showing employed adaptive and personalized interventions for the different eating behaviors

S/N	Eating Behavior (fully adaptive/personalized)/	Example Studies
1.	Eating with distractions	[11,12,18,26,27,42,57,76,82,116]
2.	Eating alone (mentioned together with eating with others)	[57]
3.	Dietary restraint	[18,51]
4.	Self-monitoring (mentioned as self-monitoring, self-tracking compliance)	[12,18,54]
5.	Slow eating	[27,64]
6.	Food awareness	[18,27]
7.	Eating speed	[64,84,116]
8.	Snacking	[96,116]
9.	Food quantity perception	[116]
10.	Healthy food choices	[84,116]
11.	Food variety (color balance, nutrient balance)	[78,84]
12.	Multisensory eating	[42]
13.	Playfulness, meal rituals	[42]
14.	Timing between meals / meal timing / meal rhythm	[11,12,78]
15.	Cravings	[12,48,96]
16.	Eating autonomy / picky eating	[57]
17.	Attentiveness to each bite / mindful eating	[18,26]
18.	Reducing meat and processed food / healthy substitutions	[69]
19.	Eating with family	[96]
20.	Sensory enhancement	[64]
21.	Chewing detection	[101]
22.	Symptom-triggered eating behavior/ingredient monitoring	[98]
23.	Food choice / dietary preferences	[113]
24.	Trying new things / dietary diversity	[48]
25.	Dietary tracking / personalized nutrition / nutrient intake visualization:	[8,97,113]
26.	Recording meals (calorie estimation, logging)	[54]
27.	Timing between drinks (drink activity logging)	[39]
28.	Sodium intake awareness and reduction	[41]
29.	Breakfast consumption	[78]

Additionally, we found that although there is a shift toward personalized and adaptive interventions, only a few studies have explicitly associated personality traits with mindful eating interventions (5 studies – 11%, Figure 10). For example, Bangamuarachchi et al. [10] passively mentioned an association between neuroticism, extroversion, and social eating, observing that neurotic individuals with high stress levels and extroverts are more prone to social eating. However, the explicit integration of personality traits in mindfulness interventions was not covered. Similarly, Kuo et

al. [69] established an association between personality traits and eating behaviors—tailoring interventions to reduce meat and processed food intake using a self-measure personality traits questionnaire—but did not conduct quantitative association analysis between personality traits and eating behaviors. Kazjon et al. [48] indirectly assessed Conscientiousness and Openness on food-related decision-making in association with eating behaviors such as cravings and trying new foods; while these traits inspired design (e.g., curiosity and dietary diversity), no formal association analysis was presented. In another related study [97], personality traits inspired and guided system design to track eating behaviors associated with mindful practices, such as monitoring nutritional intake and dietary tracking, but again, no explicit statistical analysis of trait-eating behavior association was reported. The Eat & Tell study [2] measured Big-Five personality traits to understand compliance but did not report explicit associations with specific eating behaviors. These studies highlight a potential gap in the field and present opportunities for developing the next generation of mindful eating interventions that leverage AI's adaptive capabilities to tailor interventions based on different personality traits.

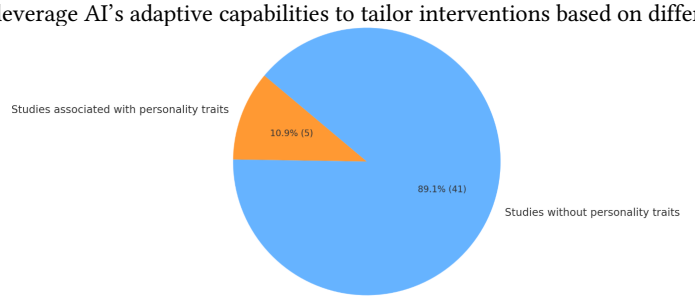

Figure 10: Distribution of studies that tailored mindful eating interventions by personality traits

3.4 RQ3: To what extent do mindful eating interventions that provide adaptive and personalized support utilize AI/machine learning techniques to accommodate different eating behaviors?

To answer RQ3, we analyzed related codes from our coding sheet. These include "*Features of AI in eating behavior technologies (e.g., ML algorithms", "sensors, Explainability and Interpretability with SHAP)", "If ML algorithm state the type (e.g., CNN)",* and *"If sensors, please state the types of sensors*".

We found that a number of AI/machine learning techniques have been employed in mindfulness eating interventions, enabling real-time sensing and personalized feedback, as seen in the studies conducted by Meegahapola et al. [81], Biel et al. [14], and Nakamura et al. [84]. Some studies also incorporate explainability features to improve transparency and trust, such as Human-in-the-loop Learning for Personalized Diet Monitoring from Unstructured Mobile Data [54]. Across these interventions, diverse AI models were employed, including Random Forest, Support Vector Machines (SVM), XGBoost, AdaBoost, logistic regression, deep learning architectures like CNNs and VGG16, and hybrid recommender systems [10,86,97,113]. These AI models are often supported by various sensors, such as accelerometers, gyroscopes, IMUs, GPS, magnetometers, piezoelectric sensors, cameras, microphones, proximity sensors, and even smart scales and pulse sensors [12,26,27,39,61,101]. This multimodal sensing enables detailed tracking of behaviors like distracted eating, slow eating, snacking, meal timing, cravings, and emotional eating in natural contexts [11,57,60,82]. Moreover, some studies emphasize personalization and context adaptation by using online learning, feature importance analysis for explainability, and real-time feedback systems [27,39,84]. This highlights a significant shift in creating transparent, adaptive, and user-centered interventions capable of delivering dynamic, individualized support to promote healthier and more mindful eating habits in everyday life. Table 11 summarizes the AI models and associated sensors used in mindful eating interventions in studies we reviewed.

Table 11: Different AI models and sensors used in mindful eating interventions

S/N	AI Models	Sensors
1.	Convolutional Neural Network (CNN) [57,113]	Magnetometer [57]
2.	Support Vector Machine (SVM) [11,62,81,86]	Step count, location, activity, WiFi, Bluetooth, proximity, app usage, notifications, screen status [62]; Accelerometer, GPS, screen usage, app usage [81]; Heart rate sensor, accelerometer (Fitbit) [86]; Accelerometer, location, battery, screen, app usage [11]
3.	Deep Neural Network (DNN) [12,81,101]	Proximity sensor, IMU (gyroscope, accelerometer) [12]; Piezoelectric sensor, accelerometer [101]; Accelerometer, GPS, screen usage, app usage [81]
4.	Logistic Regression (LR) [48]	Not applicable
5.	Naive Bayes Classifier (NB) [59]	Piezoelectric sensor, accelerometer [59]
6.	Linear Support Vector Machine (Linear SVM, LibLinear) [39]	Accelerometer, smartscale (weight sensor) [39]
7.	Random Forest (RF) [10,11,60,62,81,82,86]	Accelerometer, location, screen usage, app usage, gyroscope, step counter [10]; Step count, location, activity, WiFi, Bluetooth, proximity, app usage, notifications, screen status [62]; Accelerometer, GPS, screen usage, app usage [81]; Accelerometer, microphone, inertial motion sensors [82]; Heart rate sensor, accelerometer (Fitbit) [86]; Accelerometer, location, battery, screen, app usage [11]; Microphone [60]
8.	Naive Bayes (NB) [10,11,39,81,86]	Accelerometer, location, screen usage, app usage, gyroscope, step counter [10]; Accelerometer, GPS, screen usage, app usage [81]; Accelerometer, smartscale [39]; Heart rate sensor, accelerometer (Fitbit) [86]; Accelerometer, location, battery, screen, app usage [11]
9.	Gradient Boosting (GB) [11,81]	Accelerometer, GPS, screen usage, app usage [81]; Accelerometer, location, battery, screen, app usage [11]
10	Adaptive Boosting (AdaBoost) [10,11,81]	Accelerometer, location, screen usage, app usage, gyroscope, step counter [10]; Accelerometer, GPS, screen usage, app usage [81]; Accelerometer, location, battery, screen, app usage [11]
11.	Support Vector Classification (SVC) [10]	Accelerometer, location, screen usage, app usage, gyroscope, step counter [10]
12.	Regression Analysis (RA) [98]	None reported [98]
13.	Recurrent Neural Network (RNN) with Gated Recurrent Unit (GRU) [115]	Inertial Measurement Unit (IMU) [115]
14.	k-Nearest Neighbors (10-NN) [86]	Heart rate sensor, accelerometer (Fitbit) [86]
15.	Visual Geometry Group 16-layer model (VGG16) [84]	IMU (accelerometer/gyroscope), camera [84]
16.	Multilayer Perceptron (MLP) [62]	Step count, location, activity, WiFi, Bluetooth, proximity, app usage, notifications, screen status [62]
17.	Convolutional Neural Network (CNN) [57,113]	Magnetometer [57]

3.5 RQ4: How effective are the adaptive mindful eating interventions provided for different eating behaviors?

To answer RQ4, we analyzed the following codes from our coding sheet: *Broad category of Eating Behavior, Any mindful eating intervention provided?, Measures Outcomes, Instruments used for data collection and evaluation, Statistical tests/method of analysis, Results of empirical evaluation, Accuracy of the ML algorithm, Effective?, Description on the type of mindful eating intervention provided, and Type of intervention: adaptive/personalized or not.* Results show that out of

the 20 studies that provided mindful eating interventions, 18 were considered effective, 1 was rated as "maybe" effective, and 1 was inconclusive or not assessed (Figure 11).

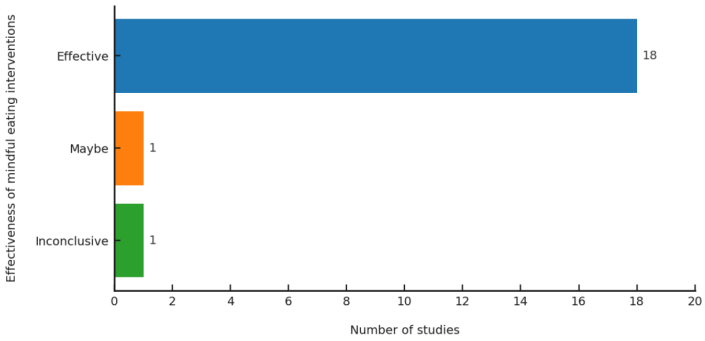

Figure 11: Distribution of the effectiveness of mindful interventions based on studies we reviewed

These results were based on measured outcomes such as eating speed, meal duration, food intake volume, food variety, emotional self-regulation, awareness scores, and symptom tracking, using a diverse range of instruments. Common tools included wearable sensors (e.g., pulse sensors [26], resistance sensors [63], magnetometers [56]); self-report surveys such as the Mindful Eating Questionnaire (MEQ) [26], Eating Disorder Examination Questionnaire (EDE-Q) [28], and User Burden Scale (UBS) [74]); photo diaries [97]; and food logs [95], machine learning models (Table 11, Section 3.2), and physiological signal tracking devices (e.g., FitNibble eyeglass wearable [12], FoodSkin edible circuits [63]).

Statistical analyses varied across studies and included ANOVA, Wilcoxon signed-rank tests, linear mixed effects models, paired t-tests, MANOVA, regression analyses, and qualitative coding techniques. For example, ViFeed reported a 15.31% reduction in eating rate and 9.65% reduction in food intake with p-values < 0.001 [27], while eat2pic showed significant increases in meal time and bite interval ($p < 0.0001$) [83]. Additionally, we assessed effectiveness in terms of AI model accuracy where applicable. AI models such as VGG16 in eat2pic achieved up to 95% accuracy [83], while models like Naive Bayes classifiers used in FitNibble reached 80% frame-level accuracy [12]. Other accuracy levels of AI models included up to 95% for intake detection, 92% for food color classification, and 93% for portion size estimation [83]; 80% frame-level accuracy for food journaling detection [12]; and an accuracy of 82.5% for sentiment-based speech classification [56]. Additionally, rule-based detection achieved 83% accuracy in identifying relevant food content interactions [28].

We also conducted a meta-analysis comparison of effect sizes emerging from seven (7) studies that explicitly reported these metrics, including Cohen's d (d), partial eta-squared (η^2), correlation coefficients (r), and coefficient of partial determination (R^2). Results of our comparison show that mindfulness eating interventions had **moderate to large** potential impacts in these selected studies, though magnitudes vary widely depending on specific design elements, measurement approaches, and targeted outcomes. For example, in the study conducted by Meegahapola et al. [80] Cohen's d values ranged from **small to large** depending on the feature analyzed, with a notable large effect for social context ($d = 0.85$). This suggests that interventions considering social context may significantly influence eating behaviors among college students. The study by Chen et al. [26] **e**xhibited exceptionally **large effects** with Cohen's d ranging from approximately *1.94 to 3.17*, indicating substantial impacts of biofeedback-based and immersive dining experiences on user engagement and mindful eating dynamics. Kuo and Horn [68] reported partial η^2 values from 0.03 to 0.10, interpreted as **small to medium effects**. This suggests that while the intervention influences health and

environmental reflections, its effect sizes on specific behavioral outcomes may be more modest. Degraen et al. [35] reported a **moderate correlation** for engagement ($r = 0.53$), equivalent to a large effect, though other outcomes were not reported. This highlights strong user engagement potential in gamified eco-feedback approaches. Mirtchouk et al. [81] reported partial R^2 values ranging from *0.05 to 0.92* for the relationship between duration and consumption. The wide range indicates variable explanatory power across different contexts and sensor-derived features, from **small to extremely large effects**. Bangamuarachchi et al.'s [11] study demonstrated **small to medium effects** (e.g., Cohen's $d = 0.12$ for top feature), suggesting limited but potentially meaningful influence of smartphone-only sensing approaches on consumption behavior detection. Aomori et al. [8] reported a **medium effect** (Cohen's d = *0.70*), indicating meaningful improvements in diet and physical activity outcomes among participants.

Effectiveness was also strongly associated with the type of intervention. Adaptive and personalized interventions (e.g., MAMAS [56], Snack Buddy [95], NutritionAvatar [40]) showed higher effectiveness rates, indicating that personalized feedback, real-time adaptation, and context-aware prompts were more capable of influencing mindful eating behaviors and supporting emotional and cognitive regulation during meals. These findings suggest that personalized interventions provide greater behavioral support and enhance engagement, leading to stronger outcomes in reducing distracted eating, supporting slow eating, and improving food awareness. However, the generalizability of these findings is limited by the exclusion criteria often applied in these studies, such as excluding participants with metabolic disorders, severe eating disorders, or psychological conditions. We expand on these limitations and other related ones in the next subsection (3.5).

3.6 RQ5: What are the limitations and design opportunities towards making mindful eating interventions adaptive to the personality of users based on the Big 5 traits?

To answer RQ5, we analyzed codes from our coding sheet, including Broad category of Eating Behavior, Personality Traits, Category of the users of the technology, Age of the users of the technology, Exclusion Criteria for participants, Challenges and limitations, Ethical issues, and any mindful eating intervention provided. We analyzed them by finding common themes based on underlying data patterns, using Braun and Clarke's [20] six-step thematic analysis procedure. More specifically, we familiarized ourselves with the coded data, generated initial codes related to intervention limitations, grouped codes into potential themes, reviewed and refined these themes to ensure coherence, defined and named each theme clearly, and finally synthesized them into a cohesive summary. Overall, we found six overarching themes that cohesively uncover the limitations of mindful eating interventions: (1) Exclusion Criteria limiting the generalizability of real-world effectiveness and scalability of mindful eating interventions, (2) Limited Integration of personality-driven design approaches, limiting the adaptive and personalized capability of mindfulness eating interventions. These themes illustrate some of the major constraints and opportunities for advancing mindful eating interventions, with potentially more inclusive, flexible, and theoretically conscious design approaches in both the field of HCI and AI.

Theme 1: Restrictive exclusion criteria and focus on adults

We found that the effectiveness of interventions can be significantly influenced by participant exclusion criteria, which may limit generalizability and future scalability. Among studies that provided interventions (i.e., those in the "Yes" category), many employed strict exclusion criteria such as excluding individuals with certain medical conditions (e.g., hypertension, diabetes, and gastrointestinal disorders), extreme BMI ranges, or mental health comorbidities. For example, studies targeting eating speed and food awareness (e.g., ViFeed [27]) or those focusing on emotional and binge eating (e.g., FoodCensor [28]) excluded participants with severe disordered eating patterns or psychological

vulnerabilities. These exclusions may improve internal validity but limit external applicability, particularly for vulnerable populations who might benefit most from mindful eating support.

Similarly, in interventions aimed at reducing meat and processed food consumption (e.g., [68]), individuals with restrictive diets or severe dietary intolerances were excluded, possibly reducing the intervention's impact on diverse dietary groups. For studies focusing on tracking and portion control behaviors (e.g., [12,74]), exclusion of participants with visual or motor impairments constrained the inclusiveness of the design. This affects the intervention's adaptability for populations with accessibility needs, an important consideration for equitable health interventions. Furthermore, some interventions for eating with distractions or emotional eating (e.g., [56,95]) excluded children with diagnosed behavioral or developmental disorders. While this ensured more controlled data, it limited insights into the effectiveness of interventions for neurodiverse populations, who may particularly struggle with these eating behaviors.

For cases where no intervention was provided (No), exclusion criteria such as age limits, health condition filters, or lifestyle restrictions still imply potential challenges for future implementation. For instance, studies exploring mood-while-eating behaviors (e.g., [10]) or multisensory and social eating contexts (e.g., [41]) may not have tested interventions yet, but future efforts to implement personalized strategies might be limited if similar exclusion criteria persist. Overall, while exclusion criteria are often necessary to protect participants and ensure study rigor, they pose limitations on the real-world effectiveness and scalability of mindful eating interventions. They constrain applicability across diverse user groups and reduce the representativeness of findings for populations with complex eating behaviors or medical needs. Future interventions should consider more inclusive criteria or adaptive designs to support broader, more equitable applications.

Additionally, most studies focused primarily on adults, with minimal inclusion of children, adolescents, or family-based contexts [27,28,65,68]. Across the dataset, the majority of interventions targeted participants aged 18 and above, with mean ages commonly ranging from the early 20s to mid-40s. For example, [27] engaged young adults with a mean age of 23.6, while [28] included participants aged 18–41 (mean = 26.7). Other studies, such as [80] and [11], focused on college students with mean ages around 23.4. While a few studies featured broader age ranges, for instance, [97], which included participants aged 18–74, or [75], which spanned ages 11 to 85 — these remained exceptions rather than the norm. Only limited interventions, like MAMAS [56] and Snack Buddy [95], explicitly engaged children or adolescents, often in caregiver-mediated contexts rather than focusing directly on younger individuals themselves.

This adult-centric focus presents significant limitations for developing truly inclusive and adaptive mindful eating interventions. Children and adolescents have different cognitive, emotional, and behavioral needs and may respond differently to mindfulness-based prompts, often requiring more playful, gamified, or family-centered strategies. Moreover, personality traits and self-regulatory capacities continue to evolve throughout development; thus, interventions exclusively designed for adults overlook opportunities for early, formative influence on healthy lifelong eating behaviors. The predominance of adult samples not only restricts the applicability of findings to broader age groups and mixed-age family contexts but also limits insights into how personality-driven adaptations might vary across developmental stages. Future work should strive for broader age inclusivity to better inform adaptive, personalized interventions that support diverse eating behaviors across the lifespan.

Theme 2: Sample, data limitations, and study durations

Other cross-cutting challenges evident in the reviewed studies were the reliance on small sample sizes, short study durations, and cultural or geographic biases, which collectively reduce the confidence in generalizing findings to broader and more diverse populations with different personality trait distributions. For example, studies like [65] and [95] engaged only 10 participants or families, while [12] included 13 participants over a short 1-week period. In contrast,

only a few studies, such as [50] (52 weeks) and [85] (3 months), featured longer-term interventions, which are crucial for assessing sustained behavioral changes.

Additionally, there was so much reliance on self-reported data in studies like [80] (70 days) and [2] (30 days), further limits the robustness of the findings. Short-term studies (e.g., [26], <1 day) also fail to capture longer-term adherence patterns and the stability of personality-driven adaptations over time. Moreover, data limitations extend to AI and ML components, though in a few studies, we highlight them to guide future work. Our findings show that inaccurate or small-scale training data restricted the ability to develop robust, generalizable models capable of adapting to diverse user behaviors [53] (Though total datasets used were not specified, the study noted this as a limitation of their work, leading to some levels of error with data collected from a few participants). Additionally, small-scale training data led to class imbalance as reported in [10]. Such methodological constraints highlight the critical need for larger, more diverse, and longer-term studies that integrate objective sensing data and personality assessments to validate and refine adaptive intervention strategies.

Theme 3: Limited integration of personality-driven design

A significant limitation identified across many studies is the lack of explicit incorporation of personality traits in the design and evaluation of mindful eating interventions. While a few studies mentioned or indirectly considered traits such as impulsiveness, self-control, or openness (e.g., [28,47,68]), the majority did not systematically assess or personalize interventions based on users' personality profiles. This omission restricts the understanding of how individual differences in personality might influence engagement, adherence, and overall effectiveness of mindful eating technologies. Personality traits, such as conscientiousness, neuroticism, and openness, are known to shape eating behaviors, decision-making, and susceptibility to external cues. Without the integration of these traits, interventions risk offering generic recommendations that may not resonate equally with all users, potentially limiting behavioral change outcomes. Moreover, the absence of personality-tailored strategies undermines the potential for developing truly adaptive and personalized systems, which are increasingly recognized as critical for long-term behavior modification and user satisfaction. Therefore, future work should prioritize the integration of validated personality assessments (e.g., Big Five Inventory, IPIP–NEO–60 [79], TIPI [45]) into mindful eating intervention frameworks. This would enable more precise targeting, improve personalization, and enhance the scalability and impact of interventions across diverse user groups.

Theme 4: Technological constraints

Technological limitations, such as sensor inaccuracies, device discomfort, and battery life issues, further complicate the real-world implementation of adaptive, in-the-moment personalized feedback based on individual traits [12,100]. For example, [12] reported user-dependent accuracy and discomfort related to wearing eyeglasses-based devices, while [100] experienced false positives during non-eating activities and lower accuracy when users wore other headgear. Similarly, [63] highlighted challenges with manual fabrication on uneven surfaces and the technical skill required for integrating edible sensors. Additional studies identified related hardware and usability constraints. In [60], battery life and misclassification due to non-eating hand movements were reported as significant barriers to continuous monitoring. Researchers such as Kalantarian et al. [59] and [58] both faced difficulties with sensor alignment and motion artifacts, leading to inaccurate or incomplete data capture. Furthermore, Flutura et al. [39] emphasized issues of battery drain and intrusive queries, which negatively influenced user willingness to engage consistently. These technological constraints hinder the seamless integration of interventions into everyday life, limiting user adherence and potentially compromising intervention efficacy. The discomfort, inconvenience, and need for frequent recharging can reduce the

appeal of such systems, particularly for long-term or daily use. Addressing these challenges is critical for advancing the development of adaptive, personality-driven interventions that can offer reliable and user-friendly support in real-world settings.

Theme 5: Ethical and privacy considerations

From an ethical perspective, privacy and autonomy concerns were consistently highlighted across the reviewed studies, especially those involving sensitive data modalities. In [74], participants expressed discomfort and social anxiety about using voice input in public spaces, raising significant concerns about voice identifiability and potential social judgment. Similarly, [12] reported participant concerns around camera-based monitoring, with users desiring control over image capture and experiencing discomfort due to a perceived loss of privacy. Additional studies emphasized these challenges. For example, [56] raised issues regarding parental stress and privacy when using audio recordings and magnetometers to monitor mealtime interactions. In [100], the use of head-mounted sensors triggered concerns about unintended data collection from nearby individuals, while [81] highlighted privacy risks from audio data and potential discomfort about being constantly recorded. Issues related to data transparency and potential misuse of deeply personal data become even more critical. As seen in [53], reliance on incremental personal data raises questions about user agencies and the long-term implications of adaptive systems. These combined ethical challenges have significant implications for user trust and willingness to engage with mindful eating interventions. Addressing these concerns through transparent data practices, robust consent processes, and clear communication about data usage is essential to support ethical deployment and widespread adoption of personality-adaptive technologies in health contexts. We present design recommendations to guide the ethical deployment of mindfulness eating interventions in future work in the next subsection (3.5.1).

3.6.1 Design Recommendations

Based on our findings on the various limitations of mindful eating interventions, we propose three design recommendations (D1 – D3). These recommendations while tailored to implementing the Big 5 personality traits for adaptive mindfulness interventions, we provide ethical guidelines on a case-by-case basis: D1: Develop Multi-layered Adaptive Interfaces Aligned with Personality Traits and User Contexts, D2: Broaden Inclusion Criteria and Incorporate Conditional Design Layers to Support Diverse Health Contexts, and D3: Integrate Ethics-by-Design Principles with Transparent Personalization Settings.

Design Recommendation 1 (D1): Develop Multi-layered Adaptive Interfaces Aligned with Personality Traits and User Contexts

To address the lack of personalization and enhance inclusivity, mindful eating interventions should implement adaptive user interfaces that dynamically adjust content and feedback based on real-time data and user personality profiles. For example, highly Conscientious users may prefer structured goals, precise progress tracking, and routine reminders [2,95,96]. Conversely, Open individuals could benefit from exploratory content, diverse sensory cues, or novel challenges [26,27]. For those high in Extraversion, incorporating social sharing, gamified group challenges, or cooperative elements can increase engagement and accountability [37,95]. Meanwhile, Neurotic users might require soothing design elements (i.e., visual and interactive elements that create a calming, low-stress environment for users), reassuring prompts (i.e., user interface messages or instructions designed to reduce anxiety and encourage continued action), and emotion-aware feedback (i.e., adapting responses or UI feedback based on detected or inferred emotional states) to reduce anxiety and prevent disengagement [28,74]. Ethical considerations to address include providing robust privacy and consent-seeking information and mechanisms, especially when collecting sensitive psychological and behavioral data (e.g., emotion

tracking, personality profiling), and flexible opt-in/opt-out settings for personalized or social features can help address autonomy and data security concerns as discussed in [12,74]. Table 12 summarizes the design elements and interaction patterns that AI technologies can advance to design personalized and adaptive mindful eating interventions in the future. AI technologies can be used to infer personality traits based on user interaction patterns, allowing for adaptive and personalized feedback. For example, repeated engagement with social features (such as sharing progress or joining group activities) may signal extraversion, prompting the system to introduce gamified social components. Similarly, irregular usage patterns, delayed responses, or emotionally charged language can indicate neurotic tendencies, guiding the interface to adopt more supportive or calming designs.

Table 12: Design elements and interaction patterns of how AI can detect social behaviours.

Personality	Design element/interaction pattern	Example	References
Conscientious	• Structured goals • Precise progress tracking • Routine reminders	• Reach a milestone within a certain period • Visual timelines showing eating habits • Push notifications	[2,95,96]
Openness to experience	• Exploratory content • Diverse sensory cues • Novel challenges	• Branching narratives • Animation and audio narration • Surprise elements	[26,27]
Extraversion	• Social sharing • gamified group challenges or cooperative elements	• Share progress with friends • Send status updates	[37,95]
Neuroticism	• Soothing design elements • Reassuring prompts • Emotion-aware feedback	• Use of soft colour tones • Onboarding messages • Chatbots	[28,74]

Design Recommendation 2 (D2): Broaden Inclusion Criteria and Incorporate Conditional Design Layers to Support Diverse Health Contexts

Many studies excluded participants with chronic diseases, mental health conditions, or dietary restrictions, limiting generalizability and excluding groups who might benefit most from personalized mindful eating support [56,95]. Future designs should adopt modular, condition-aware intervention layers, allowing adaptations for those with medical or dietary needs. For example, for users with higher *Neuroticism*, designs could integrate stress-management features and non-judgmental feedback to avoid triggering negative emotions as discussed and implied in [28]. For individuals with dietary restrictions, alternative content and culturally diverse options can be included to support both individuals who score high on Agreeableness (cooperative family or cultural eating contexts) and Openness to Experience (exploring new foods) as discussed in [68,95]. Ethical considerations to implement include detailed pre-intervention screening and dynamic risk assessment modules to mitigate potential health and ethical risks, and transparent communication on data use and intervention scope, especially for vulnerable populations, as noted in [36,106]. To implement these modifications, future mindful eating programs can adopt design elements that are personality- and condition-aware, where the design is adapted based on user personality profiles. For example:

- **User-interface personalization**: Research suggests that individuals who score high in Conscientiousness and self-control tend to maintain stable goal commitment over time, showing resistance to fluctuations in motivation and a stronger focus on long-term interests [66]. As such, they may find usefulness from a dashboard set-up to

navigate their goal-tracking, reminders, and visualizations of their progress. In contrast, considering the link between individuals who score high on openness to experience, fantasy proneness, and creativity [104], mindful eating interfaces designed for users high in Openness should include exploratory and imaginative elements such as culturally diverse meal challenges, story-based prompts (e.g., mindful eating journeys, or food stories from other times), or interactive creative tasks such as customizing meals or designing a future food scenario. These features leverage their cognitive style and are likely to sustain their engagement thought their innate curiosity about newness and imagination.

- **Nudging behaviour awareness**: AI-based systems can use sensor data (e.g., wearables), self-reports, or passive observation (e.g., patterns of mobile interactions) to detect differences in eating habits and emotional triggers [10]. For example, a rapid decline in logging food or an increase in late-night snacking could indicate stress or disordered eating patterns. With the context surrounding the situation provided by the data, the AI can suggest behaviour change strategies through psychoeducation. For example, suggested nudging [105] strategies to be used could include (a) proposing personalized coping activities (e.g., guided mindfulness or breathing exercises), (b) reminders about goals, and (c) awareness of lighter and less stressful and more nutrient-based food choices. Recent studies suggest that the influence of personality and gender on nudgeability varies by type of nudge and behavior targeted, and that tailoring nudges could be beneficial for promoting mental health help-seeking on a large scale [109]. As such, for individuals who score high on Neuroticism, the system can modulate messaging tone to be more supportive and less judgmental, whereas individuals who are low on Conscientiousness may prefer to receive simplified tracking and more frequent nudges. In this case of individuals with chronic diseases such as diabetes, the AI can further provide contextualized timing of recipe suggestions using glycemic impact to prioritize options while offering adaptive, real-time nudges for medical care goals, such as stabilizing blood sugar levels of supporting medication timing. By using personality traits, emotional state, and medical needs, the system can provide dynamic, safe, and caring nudges which will change over time based on individual pace, usage, and patterns to support behaviour regulation short-term and habit formation changes over the long-term.

Design Recommendation 3 (D3): Integrate Ethics-by-Design Principles with Transparent Personalization Settings

While integrating personality traits can greatly enhance adaptiveness, it raises critical ethical challenges related to data sensitivity, informed consent, and autonomy [12,74,100]. Future interventions should integrate ethics-by-design frameworks, making privacy protections and personalization settings core interface elements rather than secondary considerations. Key approaches should include providing clear and granular personalization controls, allowing users to adjust or disable personality-driven features, offering explicit consent stages explaining what personality data is used for and how it enhances the experience, and including feedback loops (e.g., emotion check-ins or periodic reassessment screens) that respect user autonomy and support continuous consent. Hence, ethical considerations to implement include regular ethics audits and user co-design workshops to ensure alignment with diverse user expectations and values [75] and modular privacy features, enabling different levels of personalization and data sharing depending on user comfort and context. Overall, these design recommendations will help to address the ethical complexity that will emerge while integrating further layers of personality data in mindfulness eating interventions. It is possible that the use of detailed personality profiling for personalization may intensify concerns around informed consent, as users might not fully understand how their psychological attributes are being processed and used to tailor interventions. We have extensively addressed these and other emerging ethical issues with individual design recommendations.

Additionally, systems informed by personality should accommodate potentially different ethical expectations and privacy preferences among different personality types. For example, although extroverts may generally be less

concerned about sharing behavioral data [46], it does not mean that they do not care about privacy. Extroverts can have strong opinions about how consent is conceived (and do not provide consent), as well as a clear, visual, socially contextualized explanation of how their data will be used. Yet, for more introverted individuals, privacy-aware users may wish to see uses of the data where expectations of discretion and anonymity are realized, and the data shared with as few people as possible. Variations in consent requests will require vastly different experiences. However, variations will require the design of flexible and personalized ethical-consent experiences that consider personality to support informed and sustained participation and result in dynamically flexible ethics (i.e., ethical frameworks that adapt and evolve in response to changing circumstances, societal values, and new information [29]) and trust.

4 DISCUSSION

In this section, we present the discussion of key findings emerging from our systematic review, synthesizing insights showing the effectiveness and behavioral outcomes of mindful eating interventions (4.1), user interface strategies and emerging design patterns (4.2), and limitations and opportunities for personality-driven adaptation (4.3).

4.1 Effectiveness and Behavioral Outcomes of Mindful Eating Interventions

Among the 22 studies that provided explicit mindfulness-based components, 19 were rated as effective, 2 as "maybe" effective, and 1 as inconclusive or not clearly assessed. This high effectiveness rate suggests that mindful eating interventions can positively shape dietary behaviors and enhance self-awareness during meals. Quantitatively, these interventions led to diverse measurable outcomes. For instance, Chen et al. [27] demonstrated a 15.3% reduction in eating rate and a 9.6% decrease in food intake, accompanied by improvements in meal appreciation and engagement. Similarly, Luo et al. [74] reported high levels of elaboration in speech responses (74%), reflecting increased self-reflection and reduced mindless eating tendencies. In a study conducted by Jo et al. [56], parental feedback scores improved from 83.6 to 90.1, while positive speech rates rose from 72.7% to 77.9%, indicating strengthened family mealtime dynamics and healthier child behaviors.

These effects were supported by various measurement tools, including self-report surveys (e.g., User Burden Scale [74]), behavioral logs, wearable sensor data (e.g., magnetometers and speech recognition [56]), and even advanced ML models for behavior prediction (e.g., up to 95% accuracy for intake detection [83]). Such multi-method approaches enhance the robustness of findings by combining subjective perceptions with objective behavioral data, reducing the risk of social desirability bias often seen in dietary self-reporting [16]. However, variation across studies remains a critical factor. Intervention durations varied from brief one-time sessions (e.g., [27]) to multi-week or longer-term programs (e.g., [37,68]), which likely influenced outcome sustainability and participant engagement levels. Additionally, sample characteristics were predominantly skewed towards healthy adults, often college students (e.g., [14,80]), limiting applicability to other age groups or vulnerable populations. This adult-centric focus potentially overlooks key developmental windows in children and adolescents where eating habits are established [16,107]. Moreover, technological features played an essential role in enhancing intervention effectiveness. Real-time feedback systems (e.g., visual dashboards in [35] and avatar-based coaching in [40]) promoted immediate awareness and reinforced positive behaviors. Wearable devices, such as eyeglass-based sensors in [12] and wrist-mounted IMUs in [114], allowed for passive, continuous monitoring of eating activities, reducing the reliance on self-reporting and enhancing ecological validity. Despite these advantages, wearables also introduced challenges such as comfort issues, battery limitations, and potential sensor inaccuracies [63,100]. A notable limitation across these studies was the use of strict exclusion criteria, such as omitting individuals with mental health conditions, specific dietary restrictions, or chronic diseases (e.g., [27,28]).

Such exclusions restrict generalizability and raise concerns about the scalability of interventions to real-world populations with diverse needs and comorbidities [21,32,86]. Furthermore, these criteria might disproportionately exclude those who could benefit most from mindful eating interventions, such as individuals with emotional eating patterns often linked to personality traits like higher Neuroticism or lower Conscientiousness [64].

In summary, while the majority of reviewed interventions showed strong effectiveness in improving mindful eating behaviors, the combination of self-reported and sensor-based methods, user-centered technological design, and real-time feedback emerged as critical contributors to success. However, future research should prioritize inclusivity, longer-term follow-up, and integration of diverse population needs to ensure broader applicability and sustained impact.

4.2 User Interface Strategies and Emerging Design Patterns

This subsection explores trends in user interface (UI) design and technological strategies that support mindful eating interventions. Across the studies reviewed, UI elements emerged as a central component not only for enabling self-monitoring but also for promoting engagement, fostering self-reflection, and delivering personalized experiences. One of the fundamentals of these interfaces was the incorporation of profile and login systems, which allow for secure data management and customized user experiences. Studies such as Snack Buddy [95], NutritionAvatar [40], and Eat & Tell [2] demonstrated how user profiles could store preferences, dietary restrictions, and progress data, thereby facilitating individualized prompts, adaptive feedback, and goal tracking. By centralizing user-specific data, these interfaces lay the groundwork for future personality-driven or adaptive intervention possibilities, supporting deeper personalization and long-term habit formation [91]. Activity screens and meal journaling interfaces also played a crucial role.

Many interventions included diary screens that enabled users to log meals, snacks, portion sizes, and emotional states [37,74,75]. Input modalities ranged from text and photo uploads to voice recordings (e.g., reflective audio questions [74]), enabling more nuanced and comprehensive self-monitoring. These multi-modal designs encourage greater self-awareness and accountability, which are core principles in mindful eating practice. The ability to blend structured logging with open-ended reflection supports users in identifying emotional triggers and contextual factors influencing their eating behaviors [49,101]. Real-time feedback dashboards and visual analytics were another common element, enhancing immediate user understanding and reinforcing positive behavior change. Examples include the eco-score dashboards in [35], the avatar-driven coaching feedback in [40], and the gamified progress visualization in [96]. By making abstract concepts such as nutrient balance, environmental impact, or portion control more tangible, these interfaces align closely with persuasive and HCI design principles emphasizing clear, actionable feedback loops [62]. Such visualizations not only help translate data into meaningful insights but also motivate sustained engagement and reinforce self-efficacy in dietary behavior change [70,110].

Furthermore, social and gamified features emerged as powerful tools for enhancing motivation and creating a sense of community. Interventions like Snack Buddy [95], EcoMeal [35], and Crumbs [37] included elements such as family-based comparisons, group challenges, and social sharing functionalities. These features are particularly appealing to individuals with high levels of Extraversion, who may be more motivated by external validation and collective experiences [51]. By embedding social accountability and opportunities for connection, these designs reduce feelings of isolation and provide supportive contexts for behavior change.

Additionally, we found several studies exploring state-of-the-art multisensory and immersive interfaces, incorporating innovations such as the heartbeat-driven glowing noodles [26] and the edible electronic circuits [63], demonstrating the potential of integrating tactile, visual, and even taste-related feedback. These multisensory designs can enhance emotional engagement and promote a more immersive connection with food, potentially enhancing adherence and satisfaction with interventions. The use of augmented reality (AR), virtual reality (VR), and other ambient

technologies suggest future opportunities to move beyond traditional screen-based feedback towards embodied, experiential approaches to mindful eating [52,78] with more opportunities and promising advancements, it is imperative to address several technological limitations, including sensor inaccuracies, device discomfort, and battery life constraints (as noted in [12] and [100]). These can hinder seamless integration into daily routines and may impact user adherence over time. Addressing these constraints will be critical in refining future interfaces to ensure they are both user-friendly and supportive of adaptive, real-time feedback.

In summary, the design strategies identified across these interventions highlight the importance of combining functional usability with personalized, reflective, and socially engaging features. This implies that by aligning UI patterns with behavioral science and persuasive design principles, these interventions demonstrate strong potential to enhance user commitment and support sustainable behavior change. Future work should continue to integrate personality-driven customization and address technical challenges to further optimize the impact and inclusivity of mindful eating technologies [89,108].

4.3 Limitations and Opportunities for Personality-Driven Adaptation

Based on limitations and opportunities emerging from this study, especially as it relates to the integration of personality-driven adaptive intervention, we discuss three implications of our findings. First, strict exclusion criteria, such as excluding participants with mental health conditions, chronic diseases, or specific dietary restrictions, significantly limit the generalizability and real-world applicability of the findings. For example, interventions like MAMAS [56] and Snack Buddy [95] excluded certain vulnerable populations, which may hinder the effectiveness of these interventions for individuals at higher risk of emotional or disordered eating behaviors. Such behaviors are often closely associated with higher levels of *Neuroticism* or lower levels of *Conscientiousness*, indicating that current designs may not effectively support users who are most in need of mindful eating interventions [44].

Second, a more adult-centric focus was evident across most studies, with minimal inclusion of children, adolescents, or mixed-age family contexts [27,28,65,68]. This demographic narrowness restricts the adaptability of interventions to various life stages and cultural or social contexts. Younger users and family groups may require different engagement strategies and support mechanisms tailored to their developmental stages and social dynamics [22]. By focusing predominantly on adult populations, current interventions risk overlooking critical design elements that could make them more inclusive and impactful for a wider range of users. Third, most interventions lacked explicit personality-driven tailoring despite evidence suggesting that personality traits, such as the Big Five dimensions (Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism), can strongly influence eating behavior and responsiveness to digital health interventions. The absence of adaptive features that respond to individual traits limits personalization and may reduce overall effectiveness. For instance, users high in Neuroticism may benefit from stress-reducing, non-judgmental feedback, while those with low Conscientiousness might require more structured goal setting and progress tracking [84,99].

The integration of personality data also raises substantial ethical and technological challenges. Studies involving data collection through voice inputs or wearable sensors, such as FoodScrap [74] and FitNibble [12], highlighted privacy, autonomy, and data transparency concerns. Users expressed discomfort with passive data collection and potential misuse of sensitive information, emphasizing the need for robust informed consent processes and clear communication of data use policies. These concerns are likely to intensify if interventions begin to incorporate personality profiling, requiring a careful balance between personalization and ethical responsibility [48]. Despite these limitations, several design opportunities emerge. Future interventions should adopt adaptive, condition-aware approaches that dynamically adjust content and feedback based on real-time user states, health conditions, and personality profiles. Designing for the Big

Five traits could allow for more nuanced motivational strategies, such as encouraging exploration for highly open individuals or providing social accountability for extraverted users. Furthermore, adopting an ethics-by-design framework will be essential to ensure that personalization efforts do not compromise user trust or exacerbate vulnerabilities. Features like transparent data dashboards, customizable privacy settings, and user-controlled data sharing can empower individuals to engage more confidently and securely with these technologies.

Overall, addressing these limitations through inclusive, personality-adaptive, and ethically grounded design practices, evidenced in our design recommendations D1 – D3, future interventions will offer a promising path forward. Research has shown that personality traits significantly influence eating behaviors, such as emotional eating and cognitive restraint, highlighting the importance of tailoring interventions to individual psychological profiles [4]. This aligns with our design recommendation D1, which advocates for the development of multi-layered adaptive interfaces that adjust users' personality traits and contexts, thereby enabling more personalized and effective mindful eating support [4]. Moreover, systematic reviews of mindful and intuitive eating interventions underscore the need to broaden inclusion criteria and incorporate conditional design layers to address the diversity of health contexts, populations, and cultural backgrounds [93]. This supports D2, ensuring interventions are adaptable and inclusive, particularly given the current limitations in study populations and the need for culturally tailored strategies.

Finally, as AI-driven personalized nutrition platforms become more prevalent, integrating ethics-by-design principles and ensuring transparency in personalization settings is essential to address concerns around data privacy, autonomy, and user empowerment. This underpins D3, emphasizing the necessity for transparent, auditable, and user-controllable personalization mechanisms to foster trust and safeguard individual rights in adaptive health technologies [15,67]. Based on this grounded literature, our three design recommendations will ensure that future mindful eating interventions will become more equitable, effective, and responsive to the diverse psychological and contextual needs of users [113].

5 LIMITATIONS AND DIRECTIONS FOR FUTURE WORK

Having explored 46 studies that examined mindful eating interventions across diverse technological and behavioral contexts, this review provides valuable insights into their design, implementation, and effectiveness. However, we acknowledge several limitations that guide future research directions. First, we limited our search to databases such as PubMed, ACM Digital Library, IEEE Xplore, and Scopus. Although these databases are highly reputable and cover a wide range of human-computer interaction (HCI), behavioral science, and health-related research, there are other equally important sources, including PsycINFO and Web of Science. We believe our selected databases captured significant aspects of mindful eating interventions, yet future work should incorporate additional databases to further enrich and synchronize findings across disciplines. Second, our review included only articles published in English. While this decision ensured accessibility and consistency in interpretation, it may have excluded valuable studies conducted and published in other languages. Future research should consider leveraging advanced translation methods or multilingual review teams to synthesize findings from non-English publications, thereby enhancing global inclusivity and cultural diversity in insights. Third, we focused our conceptual analysis on the Big Five personality traits. While this framework is one of the most widely adopted and validated in HCI, behavioral sciences, and psychology, other personality models (e.g., HEXACO, MBTI, or culturally specific frameworks) could provide complementary or divergent perspectives. Future studies should explore how alternative personality frameworks align or contrast with our findings and design recommendations, offering a more holistic understanding of individual differences in mindful eating behaviors.

Overall, we have addressed an important and underexplored area in literature by integrating personality considerations into the design of mindful eating interventions. Our future work will build upon the design

recommendations proposed in this review (D1 – D3) to develop adaptive, personality-driven interventions that cater to diverse eating behaviors. Specifically, we plan to leverage AI techniques, such as natural language processing and machine learning-based personality inference, to interpret users' personality traits dynamically and tailor intervention strategies accordingly. This approach promises to advance the personalization and effectiveness of mindful eating interventions, supporting more meaningful and sustainable behavior change.

6 CONCLUSION

This systematic review synthesized findings from 46 studies on mindful eating interventions, emphasizing their design strategies, effectiveness, and potential for personalization through personality-driven approaches. Our analysis revealed that most interventions demonstrated positive behavioral impacts, with 19 out of 22 mindfulness-based studies rated as effective. We identify limitations we can improve in future designs, including the predominant limitation of studies based on adult populations, strict exclusion criteria, and the lack of personality-driven tailoring. While these have restricted the generalizability and broader applicability of current interventions, our proposed design recommendations (D1 – D3), leverage the Big Five personality framework to offer a promising lens for equitable, effective, and adaptive future interventions. These recommendations point to significant design opportunities grounded in existing works, including the development of adaptive, condition-aware, and ethics-by-design interventions that explicitly incorporate personality traits. Overall, this review underscores the importance of moving beyond one-size-fits-all approaches toward more inclusive, adaptive, and evidence-based mindful eating interventions. Our proposed approach will enhance engagement and promote sustainable, healthy eating habits across diverse populations and personalities.

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A APPENDICES

In the appendix section, three levels of Appendix headings are available.

A.1 Lists of eating behaviors collected from our review studies

S/N	Eating Behaviors	Number of Studies	Selected References
1.	Eating with distractions	19	[9,11,50,56,63,65,74,75,80,81,103,115,12,14,18,26–28,37,41]
2.	Timing between meals / meal timing / meal rhythm	13	[9,11,85,97,114,12,37,50,55,58,74,77,80]
3.	Cravings	8	[9,11,12,28,47,50,80,95]
4.	Eating speed	5	[63,81,83,114,115]
5.	Snacking	4	[77,95,114,115]
6.	Dietary tracking / personalized nutrition / nutrient intake visualization	4	[8,18,75,96]
7.	Mood-while-eating / emotional eating	3	[10,74,75]

S/N	Eating Behaviors	Number of Studies	Selected References
8.	Binge eating	3	[28,80,82]
9.	Food choice / dietary preferences	3	[74,95,112]
10.	Eating with others / social eating / eating alone	2	[7,61]
11.	Food variety (color balance, nutrient balance)	2	[77,83]
12.	Multisensory eating	2	[41,63]
13.	Playfulness, meal rituals	2	[7,41]
14.	Meal description variability (granularity, composition, context)	2	[2,76]
15.	Chewing detection	2	[59,100]
16.	Symptom-triggered eating behavior / ingredient monitoring	2	[75,97]
17.	Timing between drinks (drink activity logging)	2	[14,39]
18	Recording meals (calorie estimation, logging)	1	[53]
19.	Slow eating	1	[27]
20.	Food awareness	1	[27]
21.	Food quantity perception	1	[115]
22.	Healthy food choices	1	[115]
23.	Eating with utensils (fork, chopsticks, hand)	1	[60]
24.	Portion control	1	[74]
25.	Eating autonomy / picky eating	1	[56]
26.	Overeating / undereating / eating as usual	1	[80]
27.	Attentiveness to each bite / mindful eating	1	[26]
28.	Reducing meat and processed food / healthy substitutions	1	[68]
29.	Eating with family	1	[95]
30.	Sensory enhancement	1	[7]
31.	Food sustainability awareness	1	[35]

S/N	Eating Behaviors	Number of Studies	Selected References
32.	Self-tracking compliance / food choice reasoning	1	[2]
33.	Sodium intake awareness and reduction	1	[40]
34.	Trying new things / dietary diversity	1	[47]
35.	Meal composition, portion sizes	1	[111]
36.	Breakfast consumption	1	[77]

A.2 Codes from our coding sheet showing the respective research questions that were associated with themes

S/N	Code	Description	Research Questions
1.	Bibliometric data	This covered information about the article's publication year, type of publication, venue, DOI, and author keywords.	Bibliometric analysis
2.	Broad category of Eating Behavior (e.g. fasting practices, timing between meals, cravings, eating with distractions)	Main eating behaviors targeted, such as fasting practices, timing between meals, cravings, or eating with distractions.	RQ2, RQ3, and RQ4
3.	Personality Traits	Personality dimensions considered in the study (e.g., Big Five traits) and how they relate to eating behaviors.	RQ2 and RQ5
4.	Type of technology (wearable, mobile app)	The technological platform used, including mobile apps, wearables, VR, or multimodal systems.	RQ1
5.	Components of the user interface with an explanation of their functionalities (e.g., login, activity screens)	Description of UI elements and functionalities (e.g., login, activity screens, meal diaries, feedback dashboards).	RQ1
6.	Category of the users of technology (e.g., adults, children)	The target demographic of the intervention (e.g., adults, children, families).	RQ5
7.	Age of the users of technology (e.g., 20 or age bin if not provided as a single instance)	Reported age or age bins of study participants.	RQ5
8.	Features of AI in eating behavior technologies (e.g., ML algorithms, sensors, Explainability and Interpretability with SHAP)	AI capabilities applied, including machine learning algorithms, sensor fusion, and explainability features (e.g., SHAP).	RQ3
9.	If ML algorithm state, the type (e.g., CNN)	Specific types of machine learning algorithms used (e.g., CNN, LSTM, decision tree).	RQ3, RQ4
10.	If sensors, please state the types of sensors	Sensors used in technology such as IMUs, cameras, audio, piezoelectric, or biosensors.	RQ3
11.	Name of the Eating Behavior Technology	The intervention or system name analyzed in the study.	RQ1, RQ2, RQ3, RQ4, and RQ5
12.	Data collected based on users' characteristics	Information on data collected related to user traits, context, or behaviors.	RQ3

S/N	Code	Description	Research Questions
13.	Interaction modalities (sound, haptic, gestures, visual). What kind of interaction takes place when physiological signals are extracted or when a diagnosis is provided/communicated?	Modes of interaction used (e.g., visual, auditory, haptic, gestural), especially when capturing physiological signals or delivering feedback.	RQ1 and RQ3
14.	Type of meal (e.g., breakfast)	Meal context evaluated (e.g., breakfast, lunch, dinner, snacks).	RQ1 and RQ3
15.	Any mindful eating intervention provided?	Whether a mindfulness-based eating intervention was included in the study (yes/no).	RQ5
16.	Description of the type of mindful eating intervention provided	Detailed description of the intervention type and its features.	RQ2, RQ3, and RQ4
17.	Type of intervention: Adaptive/personalized intervention (yes/no)? That is does it give generic recommendations, or it uses users' characteristics	Whether the intervention was adaptive or personalized to the user (yes/no).	RQ2 and RQ4
18.	Measures Outcomes (e.g., eating speed)	Behavioral or physiological outcomes assessed (e.g., eating speed, calorie intake, awareness levels).	RQ4
19.	Instruments used for data collection and evaluation	Tools and instruments applied to collect and evaluate data (e.g., surveys, diaries, sensors).	RQ4
20.	Statistical tests, method of analysis (e.g., ANOVA)	Analytical methods and statistical tests used to interpret data (e.g., ANOVA, regression analysis).	RQ4
21.	Results of empirical evaluation (e.g., $p < 0.001$)	Key quantitative findings, including significance values (e.g., $p < 0.001$).	RQ4
22.	Effect size (please try to calculate if the paper did not explicitly state but provided some metrics that can be used e.g. mean, standard deviation)	Reported or calculated effect sizes to assess intervention magnitude.	RQ4
23.	Accuracy of the ML algorithm (e.g., 90%)	Accuracy rates of applied machine learning models (e.g., 90%).	RQ4
24.	Effective? (e.g., yes, no or maybe)	Overall effectiveness rating of the intervention (e.g., yes, no, maybe).	RQ4
25.	Exclusion Criteria for participants (e.g. those with a history of hypertension excluded)	Participant exclusions based on health, demographics, or other conditions.	RQ5
26.	Challenges and limitations	Reported methodological or design limitations.	RQ5
27.	Number of participants	Sample size of the study population.	RQ5
28.	Study Duration	Duration of the study, including the total length of the intervention or observation period, frequency and timing of data collection points (e.g., weekly, monthly), and any follow-up assessments conducted to evaluate sustained effects.	RQ5
29.	Ethical Issues	Ethical considerations discussed, including privacy, consent, and data handling.	RQ5

A.3 Data Source (Appendix)

Our Coding Sheet (*Supplemental_File_Eating_Behaviors_Interfaces.xlsx*) showing all reviewed studies was submitted as a supplemental file for peer review.

		Bibliometric analysis						RQ1: What are
S/N	Title	Year	DOI	Type of Publication	Publication Database	Author Keywords	Geographical location	Short summary
1	Understanding	2022	https://doi.org/	Conference	ACM	Screen-based d	Multiple (Austr	A qualitative st
2	ViFeed: Promo	2025	10.1145/37065	Conference	ACM	Mindful eating;	Multiple (Singa	ViFeed strategi
3	Inferring Mood	2025	https://doi.org/	Journal	ACM	passive sensing	Multiple (Mexi	This study inve
4	Design to Eat S	2020	10.1145/34219	Conference	ACM	Eating habit int	Multiple ((Can	The paper prop
5	FoodCensor: Pr	2024	https://doi.org/	Conference	ACM	food content, ea	South Korea	The paper intro
6	Understanding	2023	10.1145/35771	Conference	ACM	smartphone sen	Multiple (UK, I	This study anal
7	BitesnBits: Infe	2018	10.1145/31611	Journal	ACM	Mobile Crowds	Switzerland	The paper prese
8	eat2pic: An Eat	2023	https://doi.org/	Journal	ACM	Behavior chang	Japan	eat2pic is an int
9	Exploring the D	2022	https://doi.org/	Journal	ACM	Human-Food In	UK	This paper pres
10	Prediction of Ea	2021	10.1145/34604	Conference	ACM	Eating activity;	Japan	The study deve
11	FitNibble: A Fi	2022	10.1145/34900	Conference	ACM	diet monitoring	United States	This study eval
12	FoodScrap: Pro	2021	https://doi.org/	Conference	ACM	Food tracking,	United States	This paper pres
13	MAMAS: Supp	2020	https://doi.org/	Journal	ACM	children; eating	South Korea	The paper prese
14	One More Bite'	2021	https://doi.org/	Journal	ACM	passive mobile	Mexico	The paper prese
15	Living Bento: E	2025	10.1145/37065	Conference	ACM	human-food int	Multiple (Japan	The study intro
16	Investigating Pr	2021	https://doi.org/	Conference	ACM	Food journaling	United States	This paper expl
17	EcoSant Lifesty	2023	https://doi.org/	Journal	ACM	Behavioral chal	United States	EcoSantÃ© is a
18	Snack Buddy: S	2015	10.1145/26751	Conference	ACM	Low socioecon	United States	This paper pres
19	FoodSkin: Fabr	2024	https://doi.org/	Conference	ACM	Human-Food In	Japan	Introduces Foo
20	MyDJ: Sensing	2022	10.1145/34911	Conference	ACM	eating detection	South Korea	MyDJ is a wear
21	Identifying and	2019	https://doi.org/	Journal	ACM	Self-tracking, p	United States	Introduces Foo
22	EcoMeal: Gam	2024	10.1145/36654	Conference	ACM	eco-feedback, e	Germany	EcoMeal is a ga
23	Yum-Me: A Pe	2017	10.1145/30726	Journal	ACM	Nutrient-based	United States	The paper prese
24	EatingTrak: De	2022	https://doi.org/	Journal	ACM	Eating detection	Multiple (USA,	EatingTrak is a
25	Recognizing Ea	2017	10.1145/31318	Journal	ACM	Nutrition; Eatin	United States	The paper prese
26	Q-Chef: The im	2022	10.1145/34911	Conference	ACM	surprise, curios	Australia	The study expl
27	Rasch-based Ta	2019	https://doi.org/	Conference	ACM	recommender s	Multiple (Germ	The paper inves
28	Eat & Tell: A R	2018	10.1145/31946	Conference	ACM	Health; Quantif	Singapore	The study evalu
29	Crumbs: Lightw	2016	10.1145/28580	Conference	ACM	Food; daily cha	United States	Food4Thought
30	Human-in-the-l	2019	https://doi.org/	Journal	ACM	Diet monitoring	United States	The paper prese
31	DrinkWatch: A	2018	10.1145/31946	Conference	ACM	Mobile Social S	Germany	DrinkWatch is
32	Recipe oriented	2022	https://doi.org/	Conference	ACM	nutrition manag	Japan	This paper prop
33	You Better Eat	2018	https://doi.org/	Conference	ACM	Virtual reality g	Australia	A two-player V
34	NutritionAvata	2019	https://doi.org/	Conference	ACM	Sodium reducti	Switzerland	The paper prese
35	Chasing Play P	2020	10.1145/33572	Conference	ACM	Human-Food In	Multiple (partic	The paper prese
36	Co-Designing E	2019	https://doi.org/	Conference	ACM	Food tracking,	United States	This study repo
37	Multimodal Fo	2018	https://doi.org/	Conference	ACM	Eating Moment	United States	This paper prop
38	Sensing Eating	2022	10.1109/ACCE	Journal	IEEE	Smartphone ser	Mexico	This study deve

39	Audio-based de	2015	10.1016/j.comp	Journal	ScienceDirect	Signal processi	United States	The paper expl
40	Monitoring eati	2015	10.1016/j.comp	Journal	ScienceDirect	Piezoelectric se	United States	This paper pres
41	Promoting Hea	2020	10.2196/17613	Journal	ScienceDirect	health promotic	Tokyo, Japan	This study eval
42	Effectiveness a	2019	10.2196/12289	Journal	ScienceDirect	remote consulta	Switzerland	The paper evalu
43	Validity and Us	2020	10.2196/16953	Journal	PubMed	mobile food rec	Canada	This study eval
44	Evaluation of tl	2021	10.3148/cjdpr-2	Journal	PubMed	Mobile applicat	Canada	The paper evalu
45	Relative Valida	2022	10.2196/40449	Journal	PubMed	dietary intake; (	Canada	The study evalu
46	Effects of a sma	2023	10.1111/jjns.12	Journal	PubMed	dyslipidemias,	Japan	This pilot RCT

e the trends in tailoring interventions for different eating behaviors?

Broad category of	Personality Traits	Type of technology	Components of the user	Category of the users of	Age of the users of the	Features of AI in eating	If ML algorithm	Datasets
eating with dist	Not explicitly s	Screen-based m	No digital inter	Adults	20-40 years	None; study wa	N/A	N/A
slow eating, foc	Not explicitly m	Mobile app-like	Live stream vid	Adults	18-29 (Mean=2	Video playback	Not a learning algorithm, but G	
Mood-while-ea	Neuroticism, E	Mobile app (iL	Notification rer	Adults (college	Mean age: 23.4	ML algorithms,	Random Forest	Multi-country c
eating speed, sn	Not explicitly m	Framework enc	Varied based on	General popula	Varied; not uni	Mention of feed	Not specified	N/A
eating with dist	Not explicitly a	Mobile app, Ch	Black cover ov	Adults with BE	18-41 (mean=2	Rule-based key	Not used	N/A
eating with oth	Not studied	Mobile app	Self-report inte	Adults (college	Mean ~24.2 yea	ML algorithms,	XGBoost, MLP	24K self-reports on
Timing betwee	Collected via T	Mobile app, W	Login, Meal/Sn	Adults (univers	18-26 years	ML algorithm u	Not explicitly named (classifica	
eating speed, fo	Not directly me	Wearable, amb	One-meal eat2p	Adults	20s	ML algorithms	VGG16	200 intake behaviors and
Multisensory ea	Not specified e	Multimodal (A	Smart tableware	Adults and chil	Children (e.g.,	Includes sensor	Not specified	N/A
Eating with ute	Not mentioned	Wearable (sma	Start/stop butto	Adults	Teen (1) and 20	ML algorithms,	Cubic SVM	274 samples of
timing between	Not directly me	Wearable (eyeg	FitNibble iOS A	Adults	21-54 (avg. 34)	ML algorithms,	Deep Neural N	FitByte DataSe
timing between	Not explicitly s	Mobile app	Meal type selec	Adults	18-60 (M=30, S	Not applied; no ML or AI features reported		
eating with dist	No	Mobile app, we	Login, meals ta	Adults (parents	2 to 8 years (ch	ML algorithm (	CNN	300k movie rev
overeating, und	Not explicitly m	Mobile app	Push notificatio	Adults (college	Mean age 23.4	ML algorithms,	Random Forest	3278 self-repor
eating with dist	Not explicitly m	Wearable (puls	Pulse sensor-en	Adults	16-65	Physiological s	None	
Meal descriptio	Not explicitly d	Multiplatform p	Login not detai	Adults	25-50 years	NLP (Nutrition	No explicit ML algorithm impl	
Reducing meat,	Conscientiousn	Mobile app	Login, daily ch	Adults	18-65	None	N/A	N/A
Snacking behav	Not explicitly s	Mobile app	Login, snack in	Adults and ado	Primary: 31-46	Not applicable	N/A	N/A
eating with distractions, sensory		Wearable, Edib	3D design supp	Adults	Average age: 2	Capacitive sensing, resistive sensing, electrical		
Chewing detect	Not reported	Wearable (eyeg	No UI compone	Adults	20-51	ML algorithm (	Deep Neural N	230 (94 eating
Meal timing, ea	Not explicitly s	Mobile app, W	Mobile app: ph	Adults	18 to 74 years	Regression ana	Regression ana	N/A
Food sustainabi	Not reported	Mobile app	Barcode scanne	Adults	22-36 years (M	None (No ML/	N/A	N/A
food preference	Not specified	mobile app, we	Survey for diet	Adults	Not specified	ML algorithms,	CNN (VGG-ba	Food-101 (cont
Timing betwee	Not studied	Wearable (wris	Not explicitly d	Adults	Mean - 22.6 (re	Deep neural ne	RNN (Gated R	No total was sp
Meal detection,	Not discussed	Wearable	Audio recorder,	Adults	20-63 years	ML algorithms,	Random Forest	No total was sp
Cravings, trying	Openness (refle	Mobile app	Login, onboard	Adults	18-24 (71%), 2	ML algorithms	Logistic Regres	Recipe1M+ dat
Nutritional inta	Self-control, su	Mobile app	Login, dietary c	Adults	18-73 (avg 27.7	Hybrid recomm	Hybrid recommender (taste and	
self-tracking co	Big-Five (via II	Mobile web app	Login, food dia	Adults	18-65+ (mostly	No ML used in	N/A	N/A
Mindfulness; ti	Not explicitly a	Mobile app	Daily challenge	Adults	18-65 (avg ~28	None (no ML or explainability		N/A
Recording mea	Not explicitly d	Mobile app	Diary screen (re	Adults	18-35	NLP, Incremen	Rule-based clas	No total was sp
timing between	Not explicitly m	Wearable (sma	Start/Stop reco	Adults	Not specified	ML algorithms,	Naive Bayes, L	5117 samples o
meal compositi	Not explicitly s	Mobile app	Recipe databas	Adults	Not explicitly p	Linked nutritio	Not specified	N/A
eating with dist	Not explicitly m	VR headset, mo	VR view (black	Adults	18-34 years (M	Sound threshol	None	N/A
Sodium intake	No	Mobile app	Login, dietary I	Adults	26-59 (Mean bi	Behavioral modeling with rule-based avatar fee		
Eating with soc	Not explicitly a	None directly d	Not applicable	Adults	Not precisely st	Not applicable	N/A	N/A
tracking food, m	Not explicitly m	Mobile app (co	Includes modul	Adults	11-85 (age bin)	None (no ML o	N/A	N/A
timing between	Not explicitly m	wearable, mobi	Daily activity s	adults	Not specified; g	ML algorithms,	Support Vector	90,720 atomic i
timing between	Not explicitly e	Mobile app	i-Log app: Entr	Adults	Mean age 23.4	ML algorithms,	Random Forest	58 college stud

Chewing, Swal	Not explored	Wearable (Sma	Android app fo	Adults	22-35	ML algorithm (	Random Forest	250
Timing betwee	Not mentioned	Wearable	Mobile app wit	Adults	22-34	ML algorithms,	Naive Bayes cl	No total was pr
Meal timing, fo	Not explicitly a	Mobile app (asl	Photo upload, r	Adults (college	19-22	Not directly me	Not applicable	N/A
Meal rhythm, c	Not specified	Mobile app	Chat with dietit	Adults	20-67 years	None explicitly	N/A	N/A
meal tracking, t	Not analyzed	Mobile app	Login, camera i	Adults	18-65+	AI-based food r	Not specified	N/A
Mindful eating,	Not specified	Mobile app	Photo-based fo	Adults	Mean age 38.5±	AI-based image	Not specified	N/A
dietary assessm	No	Mobile app	Meal photo cap	Adults (healthy	18-70	AI food recogn	Not specified	N/A
diet tracking, n	No	mobile app	Login, diet reco	adults	20-65 years, me	AI-based image	Not specified	N/A

		RQ2: How do people engage with technologies providing mindful eating interventions for different eating behaviors?						
If sensors please state	Evaluation Locations	Name of the Eating	Data collected based on users	Interaction modalities	Type of meal (e.g.	Any mindful eating	Description on the type of	Type of intervention:A
N/A	Field study (pa	N/A (naturalisti	Yes, diary logs,	Visual and audi	Breakfast, Lun	No direct interv	N/A	No
Camera	Lab (Zoom-bas	ViFeed	Eating rate, bite	Visual (video a	Lunch or dinne	Yes	Slower video p	Yes (adaptive/p
Accelerometer,	Field study	iLog	Mood, location	Visual (smartpl	Not specified (g	No	N/A	No
	Design study (l	Design Framew	Yes; personaliz	Visual, haptic,	Not meal-speci	Yes	Strategies like f	Yes (adaptive/p
None	Field study	FoodCensor	Log data (videc	Visual (black c	Not specific	Yes	Promotes reflec	No (interventio
Step count, loca	Field study (in-	Unnamed custo	Demographics,	Visual (app inte	Not specified e	No	N/A	Partially (hybri
Fitbit (step cou	Field study (rea	BitesnBits	Demographics,	Visual (photos)	All meals (brea	No	N/A	No
IMU (acceleror	Lab experiment	eat2pic	Age, interest in	Visual (digital	Japanese set me	Yes	Real-time and l	Yes
Not specified (l	Laboratory stuc	Multiple systen	Yes, includes se	Visual, haptic,	All meals (cont	Yes (indirectly)	Mindful eating	Yes
Accelerometer	Field study (nat	Not explicitly n	Acceleration da	Visual (UI with	Mixed (ramen,	No	N/A	No (Generic me
Proximity sens	Field study	FitNibble	App usage logs	Visual (notifica	Meals and snac	Yes	Just-in-time rer	Adaptive/perso
	Field study	FoodScrap	Yes, data inclu	Speech (primar	Breakfast, lunc	Yes	Guided prompt	No (generic gui
Magnetometer	Field study	MAMAS	Speech (parent-	Visual (graphs,	Breakfast, lunc	Yes	Feedback basec	Yes
Accelerometer,	Field study	i-Log mobile ap	Eating patterns,	Visual (app inte	Meal and Snac	No	Not applicable	No
Pulse sensor	Laboratory exp	Living Bento	Demographics,	Visual (glowing	Dessert/snack (	Yes	Heartbeat-drive	Yes
No physical ser	Field study (in-	ModEat	Text, voice, bar	Voice (Alexa/C	Any meal (brea	No	Mindful eating	No
None	Field study	EcoSant	Demographics,	Visual (app inte	General (includ	Yes	Daily challenge	Yes (partially p
N/A	Field study	Snack Buddy	Snack logs, phc	Visual (UI inter	Snacks	Yes	Users reflect or	Yes - feedback
Capacitive sens	Psychophysical	FoodSkin	Taste/texture di	Visual (traces),	Chocolate, cool	Yes	Monitors eating	Personalized (b
Piezoelectric se	Field study, in-	MyDJ	Eyeglass/head	Visual (wearab	Meal and snack	No	Not applicable	Yes (fine-tunin
None	Field study	Foodprint	Food photos, sy	Visual (photo r	All meals	Yes	Photo taking an	Yes
N/A	In-the-wild stuc	EcoMeal	Age, living situ	Visual (interfac	Not specific to	No	N/A	Generic, not ad
None	Field study, On	Yum-me	Dietary restricti	Visual	All meals (not l	No	Not applicable	Yes
Inertial Measur	Semi-wild field	EatingTrak	Yes (dominant	None directly; l	Snacks, meals (	No	N/A	No
Accelerometer,	Laboratory exp	ACE (Accelero	Age, gender, ac	Audio (chewing	All meal types (	No	Not applicable	Yes
None	Field study (on	Q-Chef	User taste, fami	Visual (mobile	General (not sp	No	N/A	Yes (Personaliz
None	Field study	Nutrilize	BMI, WHR, ag	Visual (nutrient	All meals (brea	No	N/A	Yes
None	Field study (rar	Eat & Tell	Demographics,	Visual (survey	Breakfast, Lun	No	N/A	No (uses generi
	Field study	Food4Thought	Yes - demograp	Visual (photo-b	All meals (chal	Yes	Non-corrective,	No (not persona
Speech-to-text	Field study (13	EZNutriPal	Unstructured sp	Visual (UI), Vc	All meals (exar	No	N/A	Yes (personaliz
Accelerometer,	Laboratory stuc	DrinkWatch	Wrist movemer	Sound (notifica	Drinks (fluid in	No	N/A	Adaptive/perso
None	Field study, usa	RecipeLog	Recipes, image	Visual	All meals (e.g.,	No explicit inte	N/A	No (generic stru
Microphone (sc	Lab-based user	You Better Eat	Age, sex, behav	Visual (crumbs	Snack (fruit-ba	No	N/A	No (not persona
dback, not ML-l	Field study	NutritionAvata	Sociodemograp	Visual (avatar a	All daily meals	Yes	Persuasive, refl	Yes
N/A	Workshop setti	N/A	Photos and desc	Visual, social, p	Various, includ	No	N/A	No (conceptual
N/A	Co-design worl	Not named (cor	Dietary problem	Visual (UI scre	All meals (brea	Yes	Encouraged sel	Adaptive/perso
Heart rate sens	field study	Personicle	heart rate, step	gesture (moven	breakfast, lunc	No	N/A	No
Accelerometer,	Field study	i-Log	Demographics,	Visual (app inp	Not specified (g	No	N/A	Yes

Microphone (au	Laboratory exp	Unnamed smar	Manual annotat	Audio (microph	Snacks (chips, a	No	Not applicable	No
Piezoelectric se	Laboratory exp	Piezoelectric se	Swallow patter	Visual (via mol	Water, Chips, S	No	Not applicable	No
None	Field study	asken	Age, BMI, cool	Visual (photo-b	All meal types (	Yes	Photo-based an	Adaptive/perso
None	Field study (rea	Oviva	Demographic, a	Visual (app inte	All meals, emp	Yes	Behavioral cou	Yes
None	Field study	Keenoa	Demographics,	Visual (photo-t	All meals (brea	No	N/A	No
None	Field study (use	Keenoa	Dietary intake,	Visual (image c	All meal types (	Yes (implicitly	Participants rep	Yes (dietitian n
	Online study	Keenoa	Yes (e.g., food	Visual (photos)	All meals (e.g.,	No		No
Step counters (	Field study (6-r	calomeal	dietary intake, p	visual (graph al	Not specified	Yes	Visualization o	Yes

RQ3: How effective are the mindful eating interventions provided for different eating behaviors?								RQ4: What ar
Measures Outcomes	Instruments used for data	Statistical tests, method	Results of empirical	Effect size (please try to	Accuracy of the ML	Effective? (e.g. yes, no or	Exclusion Criteria for	Challenges and
Chewing rate, i	Action camera,	Thematic analy	Screens influen	Not applicable	N/A	Maybe (design	None stated exp	Small sample s
Eating rate, foo	Camera (Zoom	Linear Mixed E	Eating rate redu	Not directly rep	Not applicable	Yes	Medical conditi	Food awarenes
Mood (positive	Smartphone ser	Random sampl	CBM accuracy:	Not explicitly r	63.8% (MEX),	Yes	None explicitly	Class imbalanc
Eating rate, foo	User studies, su	Qualitative ana	Framework hel	Not applicable	Not applicable	Yes (framework	Not specified	Complexity of
Food video exp	Custom logging	Friedman Test,	Significant redu	Not explicitly c	N/A (rule-basec	Yes	Under 18, not i	False positives
Social context	Smartphone ser	t-tests, Cohen's	AUC up to 0.75	Reported using	Not explicitly s	Yes (hybrid mo	Not specified	Uneven country
Eating frequenc	Mobile app, Fit	Correlation, cla	Classification accuracy = 84%		0.84	Yes	No diabetes, pr	Technical prob
Meal time, bite	Sensor-equippe	Wilcoxon signe	p < 0.0001 for l	~50% increase	Up to 95% for i	Yes	Not stated expl	Manual reportin
Eating speed, s	Interviews, PRI	Qualitative cod	Not explicitly r	Not reported	Not reported	Maybe (depend	Not consistentl	Complexity in
Eating activity	Smartwatch wi	10-fold cross-v	Max accuracy 7	Not explicitly s	Max: 76.4%, A	Maybe (modera	Not specified	Data collection
Number of mis	Surveys, app lo	Chi-square test,	19.6% improve	Not explicitly s	80% frame-leve	Yes	None explicitly	False positives,
Level of elabor	OmniTrack, Us	Descriptive stat	74% of 988 responses were elaborated; low per			Yes	Diagnosed eati	Social constrain
Mealtime lengt	Google Speech	Paired t-tests, tl	Average score improved from 8		0.825	Yes	Not explicitly s	Magnet usabilit
Food consumpt	i-Log app, sma	Pearson correla	Random Forest	Cohens-d range	87.81% (Rando	Yes	Participants wit	Self-reported da
Awareness, atte	Pulse sensor, or	MANOVA, t-te	p < 0.001 for m	Cohen's d rangi	N/A	Yes	Cold symptoms	Limited taste in
Food specificity	Daily surveys,	Logistic regress	p<0.001 (e.g., r	Not provided	NLP (Nutrition	Yes (moderate	None explicitly	Small sample,
Eating behavior	AFHC, IPAQ,	General linear	Significant imp	0.03 to 0.10 for	N/A	Yes	No smartphone	Low challenge
Snack healthine	Surveys, interv	Paired t-tests, A	t(9)=1.87, p=0.	Not stated	N/A	Maybe - qualita	Not explicitly s	Missing cultura
Eating speed, u	Multimeter, res	Wilcoxon signe	Taste distinction for 0.01mm: 65%, for 0.03mm			Yes (for electri	Not explicitly s	Manual fabrica
Eating episode	MyDJ prototyp	F1-score, precis	F1-score: 0.919	Not explicitly c	0.984	Yes	None explicitly	False positives
Awareness of e	Photo diary app	Regression ana	Photo-based da	Not explicitly s	Not reported	Yes	Not reported	Time-consumin
Interaction freq	Intrinsic Motiva	Welch's t-test,	Gamified group	r = .53 (engage	N/A	Yes	Not explicitly r	Incomplete pro
Recommendati	Web interface,	Paired t-test, W	p < 0.0001, 42.	Not explicitly r	FoodDist Top-1	Yes	None explicitly	Limited food da
Eating frequenc	Wrist-mounted	ANOVA, Preci	F1 = 73.7% (3s	Not explicitly s	Not provided di	Yes	None reported	Low detection
Eating detectio	Smartwatches,	Unpaired t-test,	TOFL accuracy	Partial R2 valu	Up to 92% with	Yes	Not explicitly s	Small sample s
Decision-makin	Mobile app dat	Student's t-test,	Significant incr	Not explicitly r	Surprise classif	Yes	Major dietary r	Limited to Wes
Optimal nutrien	Matomo tracke	Multilevel logis	Rasch-based sy	Not explicitly s	Not applicable	Yes	Minors, chronic	Short study dur
Number of com	Self-report surv	Mixed-effects l	p < 0.05 (treatm	Not explicitly p	N/A	Yes	None explicitly	Short-term stud
Mindful eating	Fred Hutchinso	Negative binom	p<0.05 for fruit	Not explicitly stated; estimate:		Yes	Non-iPhone us	Lower engagen
Accuracy of ca	Android app, U	T-test, ANOVA	Training reduce	Not explicitly r	0.897	Yes	Not specified	Default values
Drink activity r	Smartwatch (A	Simulation, cla	SVM accuracy	Not reported	Naive Bayes 76	Yes	Not explicitly s	Battery life, int
Input time, nutr	Mobile app usa	Descriptive stat	Ingredient inpu	Not provided; a	N/A	Yes, for capturi	Not stated	Longer time fo
Presence, trust,	GEQ (Game E	Descriptive stat	High immersion	Not explicitly s	N/A	Yes	None mentione	Chewing detect
Risk awareness	mFRCL food c	Paired t-test, W	Significant incr	Not explicitly reported, but mea		Yes	Not explicitly s	Small sample s
None empirical	Workshop disc	Thematic analy	N/A	N/A	N/A	N/A	None reported;	Limited cultura
Tracking comp	Paper-based co	Qualitative cod	Not applicable	N/A	N/A	Yes (formative	None mentione	Only dietitian p
Eating moment	Fitbit devices (	10-fold cross va	SVM achieved	Not explicitly s	Up to 93.33% (	Yes	None stated exp	Recognition no
Eating event de	Smartphone ser	t-tests, Cohen's	F1-score up to	Small to mediu	AUROC: 0.65	Yes	Eating disorder	Battery consum

Classification a	Samsung Galax	Leave-one-subj	F-measure: 94.4	Not reported	94.5% F-measu	Yes	Not stated	Battery life, no
Swallow detect	Piezoelectric se	Leave-one-subj	F-measure for s	Not explicitly s	Solid: 88.8%, L	Yes	Not stated	Sensor misalign
Interest level in	Online survey,	Friedman test,	P<.001 for incr	Not explicitly s	Not applicable	Yes	None stated exp	All female part
Weight, BMI, b	Questionnaires	Wilcoxon signe	Weight loss me	Medium (appro	N/A	Yes	Pregnancy, brea	No control grou
Nutrient intake,	Keenoa app, 3-	ANOVA, Pears	Significant diff	Not explicitly s	Not reported	Maybe (better a	Diagnosed dise	Portion estimat
User perception	System Usabili	Qualitative ope	Participants pre	Not reported	Not reported	Yes (based on u	Individuals with	Limited food da
Reported energ	Keenoa, ASA24	Paired t-test, W	No significant difference in energy intake. Stron			Yes	History of dieti	Potential sodiu
LDL, HDL, trig	Calomeal app,	Shapiro-Wilk, t	LDL: -4.00± 20	Cohen's d = 0.7	Not reported	Yes	On lipid medica	Small sample s

e the limitations and design opportunities towards making interventions adaptive to personality of users based on the Big								
Number of participants	Study Duration	Prospective future work	Prospective future work	Prospective future work	Prospective future work	Prospective future work	Ethical issues	Summary of key findings
10	3 weeks	Design for anti	Focus on collab	Use of discomf	Encourage inter	Help suppress c	Approved by et	Screens influen
24	1 week	Use adaptive vi	Include encour	Enable self-mo	Leverage gamif	Design with so	Ethics approval	ViFeed is a pro
678	30 days	Not explicitly s	Not explicitly s	Not explicitly s	Could benefit f	May benefit fro	Licensed under	Generic models
7	Not specified	Interactive fram	Group feedback	Goal-setting an	Social media-b	Emotion-sensit	Minimal; study	Proposed a com
22	3 weeks	Introduce custo	Leverage suppo	Use structured	Incorporate soc	Include calming	IRB approved,	FoodCensor sig
678	4 weeks	Not addressed	Not addressed	Not addressed	Not addressed	Not addressed	Privacy concer	Hybrid ML mo
122	10 weeks	Not discussed	Not discussed	Not discussed	Not discussed	Not discussed	Consent obtain	Mobile data co
50 (study 1 - 20	1 week	Use aesthetic a	Social sharing a	Routine plann	Group interacti	Gentle, non-jud	Consent, privac	eat2pic improve
18	Not specified	Design for inte	Use flavor-emo	Enable adaptive	Use of perform	Design emotio	Ethics of taste r	Multisensory a
5	1 day	Not discussed	Not discussed	Not discussed	Not discussed	Not discussed	Ethics approve	Smartwatch-bas
13	1 week	Allowing voice	Socially suppo	Refinements fo	Integration wit	Privacy-first se	Concerns about	FitNibble impro
11	1 week	Explore audio e	Design gentle,	Enable goal-set	Incorporate soc	Integrate stress-	Privacy and so	Speech input is
30 (15 parent–	3 weeks	Encourage expl	Foster positive	Emphasize rout	Enable gamifie	Minimize stress	Privacy concer	MAMAS increa
84	70 days (two c	Not specifically	Not specifically	Not specifically	Not specifically	Not specifically	Ethical approva	Food consumpt
59	<1 day (45 min	Design visually	Incorporate em	Use structured	Encourage soci	Ensure calm, c	Participants ga	Heartbeat-glow
15	2 weeks	Support journal	Account for em	Incorporate gra	Support VA-ba	Leverage stress	Privacy concer	Granularity and
139	20 days	Provide explora	Use collaborati	Incorporate hab	Include group-b	Incorporate pos	Voluntary parti	Daily behaviora
10 families	12 weeks	Include diverse	Leverage coope	Enable tracking	Enhance social	Provide reassur	Potential tensio	Snack Buddy s
10	> 1 week	Customize mul	Support for coo	Precision food	Interactive foo	Stress-reducing	Minimal; gold	FoodSkin is a v
30	1 week	Not discussed;	Not discussed	Not discussed	Not discussed	Not discussed	IRB-approved;	MyDJ achieved
11	3 - 4 weeks	Designs enabli	Enhancing coll	Tools to promo	Integrating soci	Systems to red	Concerns about	Foodprint supp
14	1 week	Study long-ter	Not discussed e	Not discussed e	Not discussed e	Not discussed e	Not discussed	EcoMeal enha
227	Not specified	Not discussed	Not discussed	Not discussed	Not discussed	Not discussed	IRB approved s	Yum-me achiev
9	113 hours	Not discussed	Not discussed	Not discussed	Not discussed	Not discussed	Not discussed	EatingTrak sign
12	72 hours	Explore openne	Investigate soci	Model predicta	Examine impac	Study stress-rel	Audio privacy	Personalized tra
42	Not specified	Modeling indiv	Use social reco	Incorporate sus	Integrate social	Enable adjustab	Participants pr	Surprising recip
98	2 weeks	Not explicitly s	Not explicitly s	Not explicitly s	Not explicitly s	Not explicitly s	Tailoring could	Tailored Rasch-
245	30 days	Explore dynam	Use cooperativ	Design routine-	Incorporate soc	Introduce emot	IRB-approved;	Random-loss in
61	3 weeks	Use creative, fl	Design group c	Tailor to struct	Leverage social	Reduce judgme	Minimal; priva	Lightweight dai
13	13 days	Use contextual/	Enhance know	Build structure	Use voice inter	Provide reassur	IRB-approved s	EZNutriPal ena
3-Jan	3 weeks	Enable customi	Improve coope	Offer progress	Enable social f	Minimize stress	User privacy, a	DrinkWatch su
16-21	Not specified	Use diverse use	Recommend co	Support for co	Integrate social	Design soothin	No direct issue	RecipeLog ena
22	Not specified	Explore broade	Enhance comfo	Regulate over-/	Support larger	Design around	None reported;	Eating real food
28	4 days	Use of AR/VR	Supportive ava	Detailed perso	Gamified elem	Emotionally su	Design caution	Avatar-based in
18	1 day	Encourage cult	Design social a	Use structured	Promote playfu	Use comforting	None reported	Playful food tra
68	Not explicit	Enable flexible	Integrate collab	Incorporate reg	Include commu	Use supportive	Data privacy, s	Customizable t
3	3 months	Expand recogn	Improve voice	Optimize journ	Incorporate soc	Track stress/em	Not explicitly d	Automated foo
58	Not specified	Leverage app u	Use passive se	Develop syste	Incorporate soc	Deliver calmin	Ethical approva	Smartphone-on

276	Not specified	Use eating sour	Improve user in	Track and prov	Integrate social	Provide reassur	None declared	Smartwatches c
30	Not specified	Use adaptive ca	Design unobtru	Enable detailed	Incorporate soc	Integrate mood	None declared	The wearable sy
38	7 days/individu	Could explore c	Could enhance	Could leverage	Could focus on	Could include s	Privacy maintai	Social media int
43	52 weeks	Not addressed	Not addressed	Not addressed	Not addressed	Not addressed	Study approved	Mobile app-bas
102	2 weeks	Not discussed	Not discussed	Not discussed	Not discussed	Not discussed	Approved by C	Keenoa app sho
72	3 days	Enable camera	Include more p	Improve portio	Add social shar	Improve interfa	Consent obtain	Users found Ke
136	2 weeks	Not specified	Not specified	Not specified	Not specified	Not specified	Approved by M	Keenoa demons
38	6 months	Not specified	Not specified	Not specified	Not specified	Not specified	Ethical approva	The interventio

ıce eating behavior aspects like pace and enjoyment. While they distract, they may also serve as companions, and their use doesn't necessarily un

mising intervention using subtle video manipulations and visual cues to promote mindful eating during screen use. It slows eating and enhances

s perform poorly in mood-while-eating context; personalization using community-based modeling significantly improves accuracy and usability i

ıprehensive design framework for eating interventions; demonstrated usability through design sessions; identified promising but underused strate

gnificantly reduced food content exposure and promoted self-awareness, reduced binge/purge symptoms, and suggested design strategies for adap

dels using smartphone data can infer eating context (alone vs with others) more effectively than time-based baselines; significant cross-country v

mbined with contextual information and ML can classify meals vs. snacks with high accuracy. Time, location, and eating patterns vary by social

ed meal duration and food variety awareness. Visual feedback effectively nudged users, especially those interested in art/games.

nd emotional aspects are key for enhancing eating experience design. Integration of internal senses and personalization are underexplored but pro

sed acceleration data can classify eating activities by food type and utensil used with moderate accuracy; hands used foods (bread, onigiri) gave b

oved compliance by 19.6%, increased snack-awareness, and reduced journaling difficulty; adaptive reminders effectively support eating behavior

effective for capturing rich, reflective food journaling data with relatively low burden, though mindful eating can be encouraged via guided pron

ased children's eating autonomy, enhanced parents' self-awareness, improved parent-child relationships, and positively influenced family mealtin

ion levels can be accurately inferred using mobile sensing; sociability, mood, and activity significantly affect consumption patterns; Random For

ing food enhances dining awareness, emotional bonding, and immersion. Sharing heartbeat enhances empathy and social connection.

l specificity vary by goal/modality; NLP interprets most entries; images show recognition challenges; journaling flexibility must match user pref

al prompts can improve both personal health and sustainability behavior. Active involvement and health framing can enhance engagement.

upported family-level healthy snacking behavior, increased awareness, facilitated engagement through tailored interfaces, but faced challenges in

iable technique for integrating edible circuits into food to monitor or augment eating behaviors via taste, heat, or feedback; found to be accessibl

l high eating detection accuracy (0.919 F1-score), strong user comfort (94.95%), and 4x battery life compared to state-of-the-art. Personalized mo

orts personalized, goal-driven dietary reflection and collaboration with health experts through photo-based tracking and visualization tools.

ıces ecological awareness; gamification boosts interaction and user motivation; app usability is high, with potential for further improvement

ved significantly higher acceptance rates than traditional survey-based meal recommenders by learning fine-grained food preferences through ada

ıificantly improves time-resolution of eating moment detection to 3 seconds using wrist-mounted IMU and arm posture estimation. Effective acr

aining data (even 1 day) significantly improves recognition accuracy; combined sensor modalities perform better than individual ones; audio alon

oes increased curiosity and exploration but reduced nostalgia and perceived confidence; personalization and contextual awareness are key for futu

-based goals improved adherence and perceived personalization and effectiveness, with gamification and nutrient prioritization aiding behavior c

ıcentives significantly increase dietary tracking compliance over fixed-loss incentives. Intervention is low-cost and robust across diverse user pro

ily food challenges can improve mindfulness and engagement. Social features enhance participation, and non-nutritional challenges improve min

ıbles accurate, user-friendly diet monitoring using unstructured data and AI-powered prompting, achieving high accuracy and user-specific learni

ccessfully demonstrated interactive machine learning on smartwatches for recognizing drink behavior. It supports adaptive modeling with user-in

bles structured food logging by recipe edits, captures household dietary variation, and reduces input burden for ingredients while maintaining acc

d in VR enhances presence, encourages cooperation, and provides a novel, immersive game experience.

ıtervention effectively increased intention and awareness to reduce salt intake, providing promising results for mHealth dietary interventions

ıditions offer rich, culturally grounded insights for designing engaging, socially meaningful Human-Food Interaction technologies, extending bey

rackers aligned with dietitians' information needs; included 32 tracking items in five categories (food, reflection, activity, symptom, physical stat

d journaling via Personicle is feasible using SVM classifiers and wearable data; accurate recognition of eating moments using physiological and

ly sensing can effectively detect eating events; personalization greatly improves accuracy; context-aware and battery-efficient models are feasibl

:an accurately classify eating sounds (chewing, swallowing) using ML algorithms, with potential for passive nutrition tracking.

ystem successfully detects swallows and classifies food type with high accuracy, and combining sensors reduces false positives from motion artif

ıteraction with dietary tracking increased students' awareness of eating habits, interest levels improved, and thoughtful decision-making regarding

;ed dietitian counseling via Oviva was feasible and effective for weight loss and eating behavior improvement in overweight and obese adults in S

ɔwed acceptable validity for several nutrients, better performance at group level, moderate usability; dietitian review improved accuracy.

:enoa,,¢ easier and more enjoyable than traditional methods, with AI features supporting mindful eating, but improvements needed in food recogr

strated moderate to strong validity compared to ASA24, higher user satisfaction, and could improve long-term adherence in dietary tracking.

n reduced LDL and improved self-efficacy, diet, and psychosocial support, suggesting mobile app and nurse-guided programs are feasible for PL

ıdermine mindfulness if harnessed thoughtfully.

awareness without reducing enjoyment.

in real-life applications.

egies like personalization and multi-stage designs.

ptive, stage-based mindful interventions.

variation necessitates diversity-aware modeling.

and environmental context.

omising.

best F1 scores.

r tracking.

mpts. Speech modality facilitates real-time awareness, though limitations exist in terms of social context, privacy, and mental load.

ne dynamics

rest models were most effective.

erences

ı cultural inclusivity and privacy dynamics.

le, safe, and effective.

odels improved performance further. It is a viable, socially-acceptable alternative for real-world eating behavior tracking.

aptive visual interactions. The system effectively addresses cold-start issues in meal recommendation with FoodDist and online learning models.

oss semi-wild and controlled studies with F1 up to 83.8%.

ne less reliable in free-living; lab-trained models moderately generalizable; meal duration not a strong proxy for intake amount

ure interventions.

change.

ofiles.

ndful eating more than nutritional ones.

ing improvements.

n-the-loop annotations and performs reasonably well using lightweight classifiers. Future directions include improving user feedback handling ar

curacy.

yond efficiency to embrace materiality, playfulness, and shared experiences.

e); highlighted the need for patient-provider collaboration and tailored tracker design.

activity data; voice command input improves usability; limitations remain in meal duration detection and accent-based NLP.

le for mobile food diaries and dietary interventions.

facts.

g food choices expanded.

Switzerland.

nition and user interface.

LWH.

nd integrating context-aware moderation.